

Neogene stratigraphy and Andean geodynamics of southern Ecuador

Dominik Hungerbühler ^a, Michael Steinmann ^b, Wilfried Winkler ^{c,*}, Diane Seward ^c,
Arturo Egüez ^d, Dawn E. Peterson ^e, Urs Helg ^f, Cliff Hammer ^g

^a Nederlandse Aardolie Maatschappij B.V. Business Unit Offshore, P.O. Box 23, 1950 AA Velsen-Noord, The Netherlands

^b Inversiones República S.A., Pasaje Los Delfines 159, Piso 8, Urb. Las Gardenias, Santiago de Surco, Lima 33, Peru

^c Geological Institute, ETH-Zürich, Sonneggstrasse 5, CH-8092 Zürich, Switzerland

^d Instituto Geológico, Escuela Politécnica Nacional, Ap. 17-1-2759, Quito, Ecuador

^e California Academy of Sciences, Golden Gate Park, San Francisco, CA, 94114-4599, USA

^f Institut de Géologie, Université de Neuchâtel, Rue Emile-Argand 11, CH-2007 Neuchâtel, Switzerland

^g Ernst Basler + Partners Ltd., Zollikerstrasse 65, CH-8702 Zollikon, Switzerland

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Abstract

The present paper reviews Tertiary volcanic and sedimentary formations in the Inter-Andean region of southern Ecuador (between 2°S and 4°20'S) in order to develop a geodynamic model of the region. The formations occur in the southern shallow prolongation of the Inter-Andean Valley between the Cordillera Real to the east, and the Cordillera Occidental and Amotape–Tahuín Provinces to the west. One hundred fifty zircon fission-track analyses has established a detailed chronostratigraphy for the sedimentary and volcanic formations and several small intrusions. The Paleogene to early Miocene formations are dominated by intermediate and acidic volcanic and pyroclastic rocks. In addition, relics of Eocene continental sedimentary series have been identified.

The Neogene sedimentary series lie unconformably on deformed and eroded metamorphic, sedimentary and volcanic formations. They were deposited in two stages, which are separated by a major unconformity dated at ≈ 10 –9 Ma. (1) During the middle and early late Miocene (≈ 15 –10 Ma) marginal marine deltaic, lagoonal, lacustrine and fluvial environments prevailed, which we group under the heading “Pacific Coastal sequences”. They presumably covered a greater surface area in southern Ecuador than their present occurrence in small topographic depressions. We suggest that they were deposited in the shallow marine Cuenca and Loja Embayments. Deposition in a marginal marine environment is also supported by the occurrence of brackish water ostracods and other fauna. (2) Above the regional (angular) unconformity, the coastal facies are overlain by late Miocene (≈ 9 –5 Ma) continental alluvial fan and fluvial facies which are in turn covered by mainly airborne volcanic material. They represent the “Intermontane sequences” of the basins of Cuenca, Girón–Santa Isabel, Nabón, Loja and Malacatos–Vilcabamba.

Sedimentologic and stratigraphic results are used to discuss the tectonic setting of Neogene sedimentation in the forearc and arc domain of the Ecuadorian subduction system. During the Pacific Coastal stage, northward displacement of the coastal forearc block along the Calacali–Pallatanga fault zone has driven crustal collapse in the Inter-Andean region. As a

* Corresponding author. Fax: +41-1-632-1080.

E-mail address: wilfried.winkler@erdw.ethz.ch (W. Winkler).

result, extensional subsidence drove the eastward ingression of shallow seas into the Cuenca and Loja Embayments from the Manabí and Progreso Basins to the west. Tectonic inversion in the forearc area during the early late Miocene (at ≈ 9.5 Ma) reflects the initiation of W–E oriented compression and uplift in the Inter-Andean region and the establishment of smaller Intermontane stage basins, which host the continental sequences. Coeval topographic rise of the Cordillera Occidental is indicated by the onset of clastic input from the west. The small Intermontane Basin of Nabón (≈ 8.5 – 7.9 Ma) formed during the period of maximum compression.

The present data prove that the Neogene Andean forearc and arc area in southern Ecuador was a site of important but variable tectonic activity, which was presumably driven by the collision and coupling of the Carnegie Ridge with the Ecuadorian margin since ≈ 15 – 9 Ma. © 2002 Elsevier Science B.V. All rights reserved.

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1. Introduction

The Ecuadorian Andes are a component of the Northern Andes segment north of the Huancabamba deflection (Gansser, 1973). Repeated accretion of oceanic and arc elements during Mesozoic and Cenozoic times distinguishes them from the Central Andes south of the Huancabamba deflection, which did not experience a history of terrane accretion. In the Ecuadorian Andes, early Cretaceous to Recent convergence between the Pacific oceanic plates and the South American continental plate has given rise to a series of tectono-stratigraphic units, which were accreted at successive times (Fig. 1A). From east to west these are: (1) several Palaeozoic to Cretaceous metamorphic and volcanic terranes of the Cordillera Real, which accreted during the early Cretaceous (at ≈ 140 – 120 Ma; Litherland et al., 1994; Spikings et al., 2000, 2001); (2) the volcanic Pallatanga Terrane, accreted during Campanian–Maastrichtian (Hughes and Pilatasig, 1999); and (3) the Paleocene–Eocene volcanic arc Macuchi Terrane and the Cretaceous Piñon Terrane (Costa), which accreted during the Eocene (Egüez et al., 1988; Daly, 1989; Bourgois et al., 1990; Jaillard et al., 1995). The Macuchi and Pallatanga Terranes constitute the present Cordillera Occidental and are sutured to the east against older accretionary complexes which underlie the Inter-Andean Valley (Aspden et al. 1995). The proximal Amazon Foreland Basin to the east of the Cordillera Real is composed of steeply dipping thrust slices and two frontal foothill highs (Napo and Cutucú antiforms), which are commonly referred to as the Sub-Andean Zone (Fig. 1A). Within the Sub-Andean Zone, conspicuous tectonic uplift is evident exposing

Paleozoic to Tertiary basement, cover, and early foreland basin formations. A major fault separates the Sub-Andean Zone from flat-lying late Cretaceous–Recent sedimentary rocks of the foreland basin, referred to as the Oriente region (Fig. 1A).

Several middle Miocene to Pliocene clastic sediment series and volcanic sequences exposed in the Inter-Andean region (Fig. 1B) are thought to closely reflect the younger tectonic history of Neogene arc of the southern Ecuadorian Andes (e.g. Noblet et al., 1988; Hungerbühler et al., 1995; Hungerbühler, 1997; Marocco et al. 1995; Winkler et al., 1996; Steinmann, 1997; Steinmann et al., 1999; Deniaud, 2000). These sequences were attributed to the Cuenca, Girón–Santa Isabel, Loja Malacatos–Vilcabamba and Nabón Basins. The sediment series in the Catamayo and Gonzanamá areas were only recently recognized to be of Miocene age (Hungerbühler, 1997; Steinmann et al., 1999). The middle Miocene sedimentary formations most likely were laid down over a greater surface area in southern Ecuador although later stages (late Miocene–Pliocene) of sedimentation were limited to the larger perimeters of the present outcrops of the basins as shown in Fig. 1B. The Miocene basins occupied an interarc position and formed coevally with the Progreso, Manabí and Borbón basins (Baldock, 1982) in the forearc to the west, and the retroarc basin of the present day Sub-Andean zone and Oriente to the east (Tschoopp, 1953).

We present a lithostratigraphic, sedimentologic and chronostratigraphic compilation spanning from the Eocene onward, which has been acquired during several projects since 1991. The presence of numerous volcanic formations and pyroclastic intercalations in sedimentary formations favoured the dating

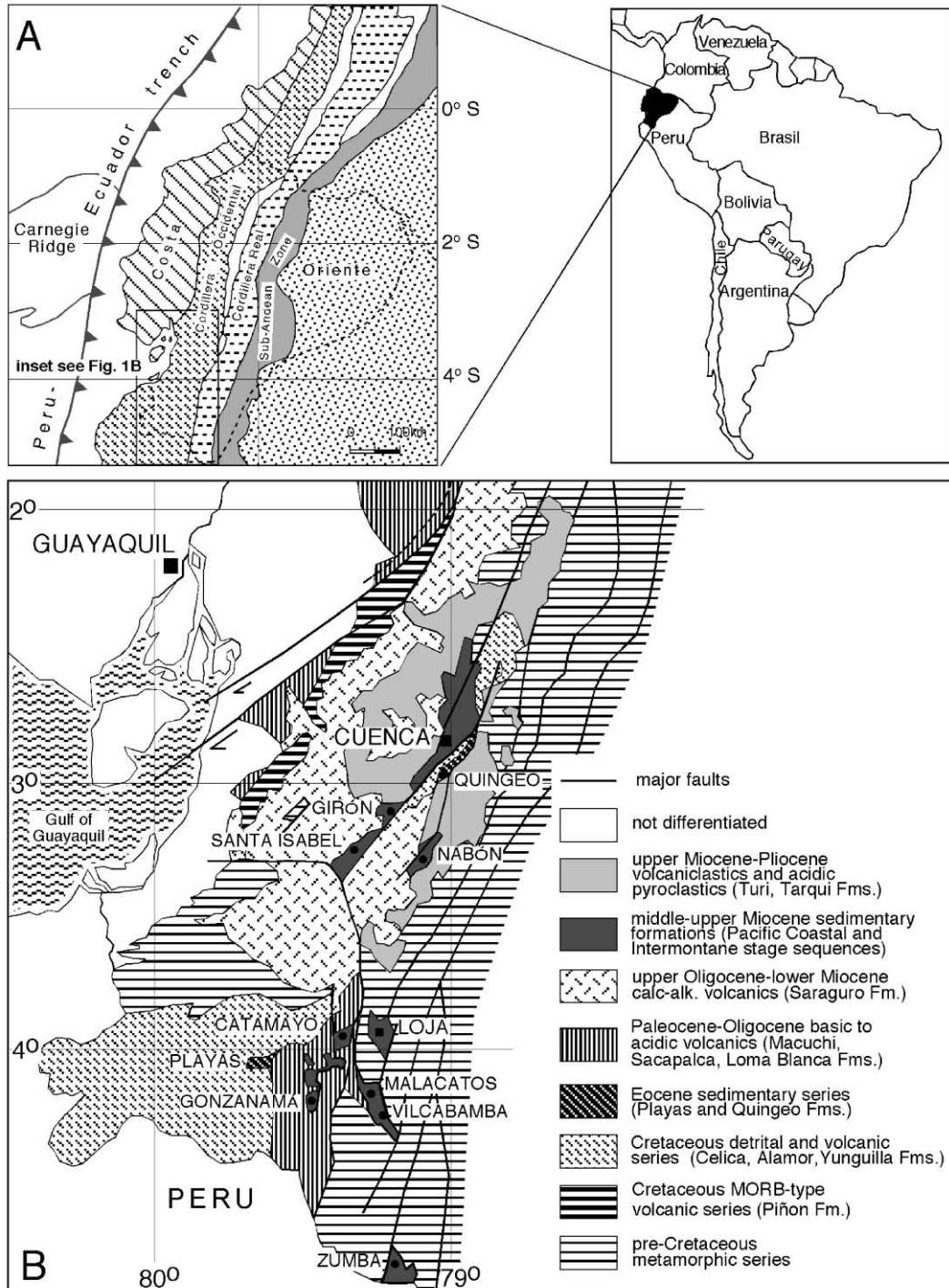


Fig. 1. Simplified maps of Ecuador. (A) Morphotectonic subdivision of Ecuador. (B) Geological map of southern Ecuador with the location of the Miocene sedimentary series. Modified from Litherland et al. (1993).

of sedimentary formations using the zircon fission-track (ZFT) method. For methods used in the fission track studies, see Hungerbühler et al. (1995), Hungerbühler (1997), Steinmann (1997) and Steinmann et al. (1999). The fission-track ages are compiled in Appendix A and are shown with their associated 2σ errors in the various figures. We also present data derived from ostracod analyses of middle Miocene sediments (Peterson et al., 2002). Finally, the data is integrated into a regional tectono-sedimentary model of the southern Ecuadorian Andes.

2. Stratigraphy and facies

In this section the characteristics of the volcanic and sedimentary formations are catalogued using a combination of lithology, stratigraphic age and the region of occurrence. The first time a formation is referred to, it is underlined, and members are printed in *italics>, and where lithostratigraphic nomenclature differs from previous work, UTM grid references of type sections and localities are provided. A simplified stratigraphic scheme is presented in Fig. 2 and chronostratigraphic correlations follow Berggren et*

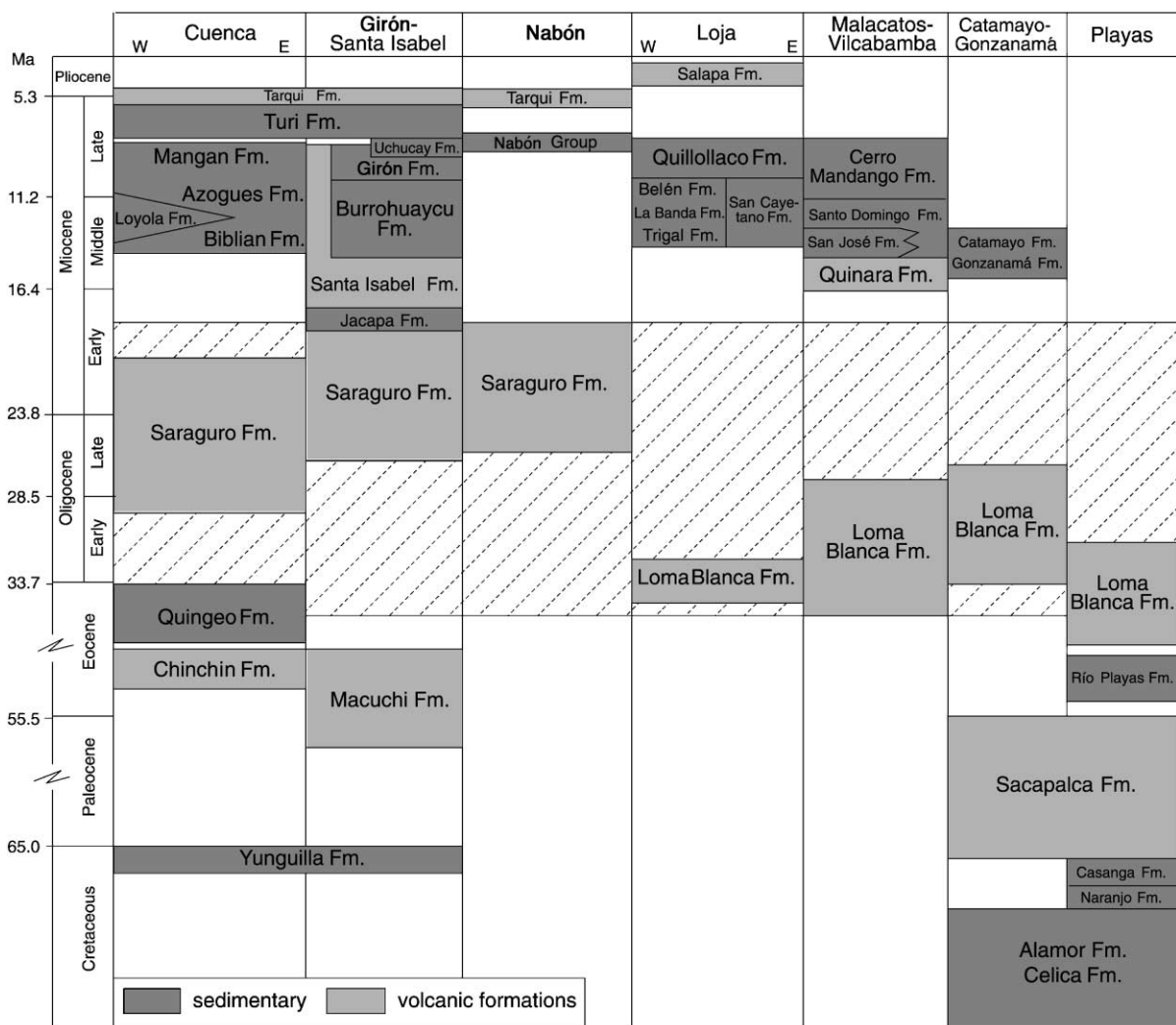


Fig. 2. Guide to the stratigraphic terminology applied in the present paper. The hatched box indicates the known chronostratigraphic range of the Saraguro Group (Dunkley and Gaibor, 1997a). For other references, see text.

al. (1995). In Appendix A the fission-track ages are compiled. Note that the samples are listed after their geographical occurrence. In case of the presence of several age components in a volcanic sample, the youngest grain population is taken to represent the time of formation.

2.1. Volcanic formations and intrusions

The Tertiary Inter-Andean sedimentary series of southern Ecuador are bounded by various volcanic formations (Fig. 2), which have been dated in several places (e.g. Hungerbühler et al., 1995; Winkler et al., 1996; Steinmann, 1997; Steinmann et al., 1999; Hungerbühler, 1997). ZFT ages from the volcanic rocks partly confirm K/Ar ages from various mineral phases (plagioclase, hornblende, biotite) and whole rocks (e.g. Kennerley 1973, 1980; Baldock, 1982; Barberi et al., 1988; Lavenue et al. 1992; Litherland et al., 1993). The following section presents the main volcanic formations, although several local ones are described in the chapters that discuss selected areas.

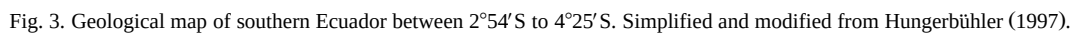
2.1.1. Paleocene–middle Miocene

The Sacapalca Formation, first described by Kennerley (1973), is up to 2000 m thick (Baldock, 1982) and is exposed in a N–S-trending belt (100 km length by 30 km width) between the towns of San Lucas and Cariamanga (Fig. 3). It is comprised of andesitic to dacitic tuffs, lava flows and volcanic breccias, which lie unconformably on Paleozoic metamorphic rocks of the Cordillera Real and the El Oro Province and on the sedimentary Celica, Alamo, Naranjo and Casanga Fms. (Kennerley, 1973; Jaillard et al., 1996, 1999) (Fig. 2). A dacite yielded a ZFT age of 66.9 ± 5.8 Ma (latest Maastrichtian to earliest Paleocene) which suggests a late Campanian to Maastrichtian (?) age for the underlying Casanga Fm. (Jaillard et al., 1996) and a middle Eocene to late Oligocene age for the overlying Loma Blanca Fm. (see below). Several plutons intruding the Sacapalca Fm. have yielded a spread of K/Ar ages (hornblende, biotite, whole rock) spanning 70 ± 2.0 to 21.2 ± 0.6 Ma (Kennerley, 1980; Herbert and Pichler, 1983; Aspden et al. 1992; Jaillard et al., 1996). We have made additional ZFT ages for the San Lucas Pluton (39.1 ± 3.0 Ma; Steinmann,

1997) and the El Tingo Pluton (21.2 ± 2.6 Ma; Hungerbühler, 1997) (Appendix A). The newly named Rodanejo Pluton yielded a ZFT age of 38.7 ± 5.6 Ma (Hungerbühler, 1997) (Appendix A). The variations in ages may be partly explained by different closure temperatures of the mineral phase and variable post-crystallisation cooling histories.

The Chinchín Fm. (Fig. 2) is exposed over a large area between Quingeo and Gualaceo (Fig. 4) and was named by Steinmann (1997) after the local village of Chinchín (type locality 740000/9681000). It was previously mapped as part of the late Miocene Tarqui Fm. (see below), which unconformably covers the Chinchín Fm. The Chinchín Fm. is composed of a very thick (maximum 3500–4000 m) succession of basic and intermediate volcanics, lava flows (partly pillowed) and minor airfall tuffs. In the upper part of the formation, aquatic reworking of the volcanics is observed. A single ZFT age of 42.8 ± 3.8 Ma (middle Eocene) was obtained (Steinmann, 1997) (Appendix A) from an andesite in the upper third of the formation. If the base of the Saraguro Group (see below) is of middle Eocene age, the Chinchín Fm. may represent an early unit of the Saraguro volcanic event.

The widespread upper Tertiary volcanic formations were mapped and described by various authors (Kennerley, 1973, 1980; Baldock, 1982; Litherland et al., 1993; Steinmann, 1997; Hungerbühler, 1997) without defining a tight stratigraphic terminology. Recently, these calc-alkaline volcanics were redefined in the Cordillera Occidental as the Saraguro Group (Kennerley, 1973) by Dunkley and Gaibor (1997a,b) (see also Pratt et al., 1997b; McCourt and Duque, 1997) (Fig. 2). According to these authors, the group contains a great number of formations and informal units and comprises intermediate and acidic subaerial volcanic rocks of late middle Eocene to early Miocene age. The base of the entire volcanic sequence has not been satisfactorily dated and could extend into the middle Eocene (J. Aspden, personal communication, 2001). These volcanic formations are not the main object of the present paper and we apply a broader stratigraphic nomenclature by considering the Loma Blanca and Saraguro Formations (described below) as parts of the Saraguro Group. The new age data will help in future to refine the stratigraphy.



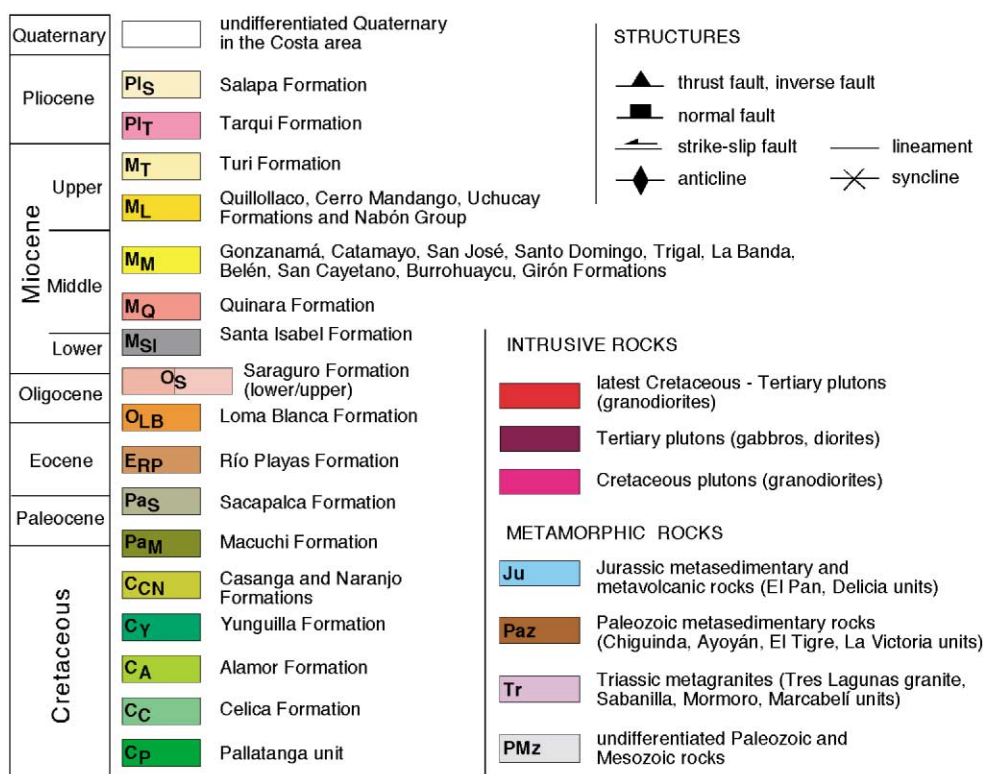


Fig. 3 (continued).

The up to 2000 m thick *Loma Blanca Fm.* (Kennerley, 1973, 1980) occurs in two main areas; one between Catamayo and Malacatos, the other N and E of Catacocha (Fig. 3). In addition, some smaller occurrences (e.g. south of Loja) are attributed to the formation by chronstratigraphic and lithologic correlations (Hungerbühler, 1997). The *Loma Blanca Fm.* consists of intermediate to acidic pyroclastics, i.e. mainly ignimbrites, pumice crystal tuffs and volcanic breccias. Dykes and sills indicating the proximity of the main eruptive center frequently intrude them. Several local occurrences of aquatically reworked intervals are observed; the thicker (100 m) *Solanda Mb.* (type locality 686260/9537030) in the area of the Río Solanda was so named by Hungerbühler (1997). The *Solanda Mb.* is characterized by metamorphic pebble-bearing conglomerates, red and green sandstones and shales deposited by mixed-load rivers, and minor tuff inter-

calations. Quartz porphyries west of the Río Solanda were described by Kennerley (1973) and are assigned to the *Purunuma Quartz Porphyry Mb.* The age of the *Loma Blanca Fm.* was constrained by 11 ZFT measurements on ignimbrites and tuffs (Hungerbühler, 1997; Appendix A). In the area of Malacatos–Vilcabamba four samples yielded a large spread of ages (40.6 ± 5.4 – 26.6 ± 4.0 Ma). A small outcrop south of Loja yielded an age of 36.2 ± 6.8 Ma and two samples collected above the Río Playas Fm. yielded 42.2 ± 3.4 and 31.1 ± 2.8 Ma. A volcanic plug located NW of the town of Catamayo (Fig. 3) gave a ZFT age of 25.2 ± 3.2 Ma, and an ignimbrite NW of Santa Rita yielded a similar age of 29.0 ± 2.8 Ma. Finally, a tuff in the *Solanda Mb.* yielded a ZFT age of 36.5 ± 4.4 Ma, and the *Purunuma Quartz Porphyry Mb.* in the area of the Loma Riodopamba gave an age of 30.3 ± 2.4 Ma. The *Loma Blanca Fm.* was often confused with the

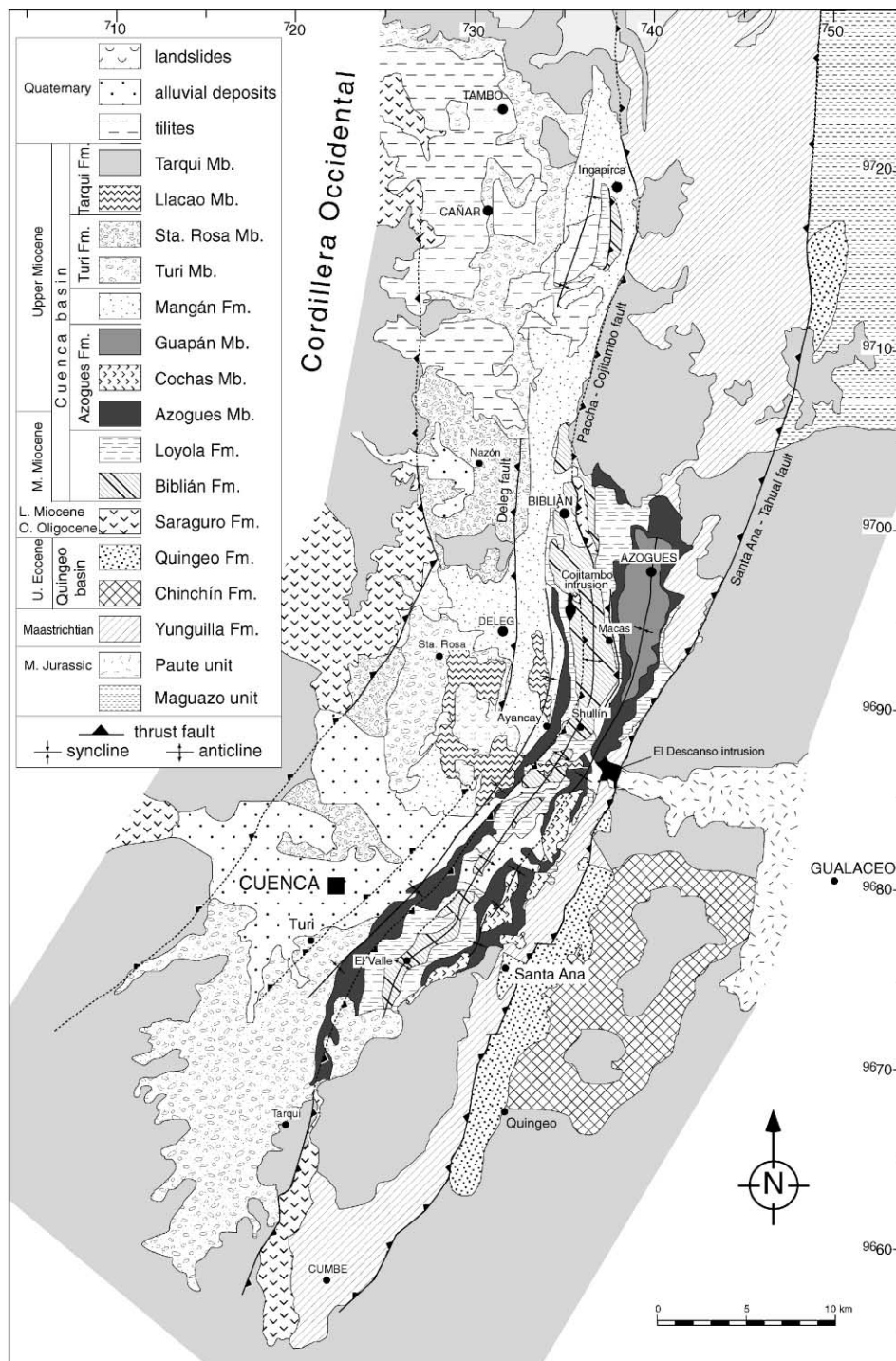


Fig. 4. Geological map of the Cuenca area between Tambo and Cumbe. From Steinmann (1997).

younger Saraguro Fm., but despite of lithologic similarities, they were clearly erupted/deposited at different times and in different geographical regions.

The Saraguro Fm., as defined by Kennerley (1980) and mapped by Baldock (1982), is the volcanic formation of greatest areal extent in southern Ecuador (Fig. 1). Its exposure spans a distance of 220 km from the town of Riobamba in the north to Saraguro/Oña in the south, and it crosses from the Cordillera Occidental in the west to the Cordillera Real in the east. The 500–2000-m-thick Saraguro Fm. unconformably overlies a series of older formations: (1) the Yunguilla, Chinchín and Quingeo Fms. in the Cuenca area; (2) the metamorphic Jurassic–early Cretaceous Alao–Paute Terrane and the Triassic Tres Lagunas Granite of the Cordillera Real; and (3) Cretaceous to Paleogene volcanics of the Cordillera Occidental (Macuchi, Pallatanga, Sacapalca and Celica Fms.) (Figs. 2–4). Several ambiguities existed concerning stratigraphic and regional correlations. However, mapping and geochronological results show that the distinction between “Pisayambo volcanics” and the Chinchillo Fm. (Litherland et al., 1993) is not necessary because lithologies and ages of these volcanic rocks integrate well into the Saraguro Fm. as defined by Kennerley (1980) (see also Pratt et al., 1997a). Similarly, the interpretation that the Saraguro Fm. is younger than the middle Miocene Burrohuyacu Fm. in the Santa Isabel–Girón area (see below) was misleading and was based on the assumption that these volcanics stratigraphically overlie the Miocene sediments (Baudino et al., 1994). However, vertical stacking is now interpreted to be a result of northwestward-directed thrust faulting superposing the older volcanics of the Saraguro Fm. onto the Burrohuyacu Fm. (Fig. 3, Hungerbühler, 1997; Pratt et al., 1997b; Hammer, 1998).

Variable deposition of the volcanic material over the pre-existing topography resulted in the thickness of the Saraguro Fm. varying between 500 and 2000 m. The Saraguro Fm. consists of intermediate to acidic pyroclastics. In the lower part, andesitic to dacitic tuffs and lava flows prevail. The upper Saraguro Fm. (Fig. 3) contains predominantly rhyolitic ignimbrite horizons of great areal extent, which show typical columnar cooling features and occasionally pumice fiamme-welding textures. Coarse

co-ignimbrite breccias and sub-volcanic rocks reflect the near proximity of the eruption centers. Inter-calated fluvial and lacustrine sediments are frequent and record periods of aquatic reworking between eruptive stages. Steinmann (1997) proposes that these voluminous ignimbrite flows were supplied by fissure eruptions and caldera-forming processes, which formed in an extensional tectonic setting.

Many ZFT ages have been acquired from the Saraguro Fm. From the Nabón area four age determinations range between 26.4 ± 4.5 and 19.0 ± 3.5 Ma (Hungerbühler et al. 1995). Fourteen samples from the Cuenca region range between 29.4 ± 2.6 and 20.5 ± 2.0 Ma (Steinmann, 1997), and six samples between Santa Isabel and Oña yield ZFT ages between 26.4 ± 2.6 and 19.1 ± 1.4 Ma (Hungerbühler, 1997). These ages correlate with a late Oligocene to early Miocene age for the Saraguro Fm. and corroborate with several K/Ar ages (biotite, plagioclase, and whole rock) reported by Kennerley (1980), Barberi et al. (1988) and Rivera et al. (1992). Lavenue et al. (1992) reported two K/Ar ages in the range of ≈ 35 Ma. However, there are doubts about their sample locations and the samples may belong to the Huigra Tandapi unit or the El Descanso Andesite intrusion (Figs. 11 and 13; Egüez et al., 1988).

To resolve the complicated stratigraphic and sedimentary relationships in the Girón–Santa Isabel area, Hungerbühler (1997) and Helg (1997) introduced the new term Santa Isabel Fm. (type locality 689714/9631727). The largest continuous outcrops occur between the towns of Santa Isabel and Girón (Fig. 3) in a north–south-trending belt, parallel to the middle Miocene sedimentary series of the Girón–Santa Isabel area to the east. This volcanic succession was previously assigned to the Saraguro Fm. (Baudino et al., 1994), the Tarqui Fm. (Randel and Lozada, 1974) and the “Pisayambo volcanics” (Litherland et al., 1993). However, in our present interpretation, the Santa Isabel Fm. is identical to the Santa Isabel Andesite Fm. of Pratt et al. (1997a,b). The formation unconformably overlies the Saraguro and Jacapa Fms. and is partly unconformably overlain by the Burrohuyacu Fm. and partly contemporaneous with the Burrohuyacu Fm. (see also below and Fig. 9). The Santa Isabel Fm. is composed of intermediate lava flows, volcanic breccias and minor tuff beds and displays highly variable thicknesses of 500–1500 m

due to volcanic infilling of pre-existing topography. Six ZFT ages yielded ages between 18.8 ± 2.2 and 8.0 ± 2.2 Ma (Hungerbühler, 1997) (Appendix A).

2.1.2. Late Miocene

The *Tarqui Fm.* is the most widespread late Miocene volcanic series in southern Ecuador. In our stratigraphic scheme (Fig. 2), it also includes several local volcanic deposits that, because of similar age and characteristic lithological features, can be classed as members of the *Tarqui Fm.* (e.g. *Tarqui*, *Llacao* and *Tambo Viejo* Members of Steinmann, 1997 and Hungerbühler, 1997). With the exception of the local *Salapa Fm.* (described below), Pliocene volcanics are generally not observed in southern Ecuador.

The intermediate to acidic pyroclastics of the *Tarqui Fm.* (Bristow, 1973) cover large areas in southern Ecuador (Figs. 1, 3 and 4); its exposed volcanic ashes are often altered to dark red and purple kaolinitic clays. The *Tarqui Fm.* unconformably overlies a wide range of Tertiary sedimentary and volcanic formations (e.g. the *Saraguro* Group, the late Miocene *Nabón* Group, middle to late Miocene formations in the *Cuenca* region, see below). The formation hosts a large variety of lithologies including rhyolitic to andesitic volcanic breccias, ashflow tuffs, pyroclastic flows, ignimbrites and many airborne tuffs (Baldock, 1982). Steinmann (1997) distinguished two members in the *Cuenca* area (Fig. 4): (1) the *Tarqui Mb.* (type locality 718800/9667000), which consists entirely of poorly consolidated and deeply weathered red volcanic airfall deposits of ≈ 300 m thickness. Eight samples revealed ZFT ages between 6.8 ± 0.8 and 5.5 ± 0.6 Ma (Appendix A; Steinmann, 1997). (2) the *Llacao Mb.* forms the entire plateau of *Loma Cochamama* (730000/9687800, Fig. 4) SW of *Azogues*, where it overlies the *Mangán Fm.* with a pronounced angular unconformity (60°). The *Llacao Mb.* mainly represents deposits of a volcanoclastic alluvial fan with channel fill and overbank sediments, as well as debris flows derived from a western source. Intercalated airfall deposits are rare. A ZFT age of 5.1 ± 0.6 Ma from the young part of the *Llacao Mb.* was obtained (Steinmann, 1997; Appendix A). The *Tambo Viejo Mb.* in the *Nabón* area (*Tambo Viejo Fm.* in Hungerbühler et al., 1995) also

mainly consists of red airborne volcanics and gave a ZFT age of 6.3 ± 1.0 Ma (Hungerbühler et al., 1995).

Several late Miocene intrusions are present in the *Cuenca* area: (1) The light grey dioritic *Cojitambo* intrusion forms the prominent peak of *Cojitambo* SW of *Azogues* (Fig. 4). Radial columnar cooling structures suggest that the intrusion penetrated the sediments at shallow depth. The intrusion cuts the tectonically deformed, middle to late Miocene sedimentary series of the *Cuenca* area (Steinmann et al., 1999). Two ZFT ages of 5.4 ± 0.6 and 7.8 ± 0.8 Ma have been obtained from the intrusion (Appendix A; Steinmann, 1997). The latter age is in good accordance with an Ar/Ar age of 7.5 ± 0.44 Ma (single crystal plagioclase, Madden, personal communication, 1996). The younger 5.4 ± 0.6 Ma fission-track age, obtained from a large dacite block displaying flow structures located SW of *Cojitambo* (737503/9691772), may suggest a later extrusive phase. Several K/Ar ages (plagioclase, whole rock) reported by previous authors range from 7.1 ± 0.3 and 6.3 ± 0.2 to 5.2 ± 0.2 Ma (Olade, 1980; Barberi et al., 1988; Lavenue et al., 1992). (2) Several intrusions have been observed south of *Quingeo*, in the region of the *Loma Chimborazo* and *Loma Gualashi*. The intrusion near the *Loma Gualashi* yielded a ZFT age of 7.6 ± 1.0 Ma (Appendix A; Steinmann, 1997), suggesting that intrusion was coeval with the *Cojitambo* event.

2.2. Sedimentary formations

2.2.1. Río Playas and Quingeo formations

Several earlier workers assigned these formations to the Miocene (Kennerley, 1980; Noblet et al., 1988; Marocco et al., 1995). However, new chronostratigraphic data suggest that they represent relics of older and more widespread sedimentary successions, which may provide clues to the Paleogene history of the Ecuadorian Andean chain.

The *Río Playas Fm.* (Kennerley et al., 1973) occurs in a relatively small, 15 km long by 6 km wide, depression between the villages of *Casanga* and *San Antonio* (Hungerbühler, 1997; Fig. 3). In the north the thickness reaches 500 m and it pinches out to a few tens of meters in the south. The present definition of the formation follows the suggestion of Jallard et al. (1996), who restricted the formation to the

yellow coloured conglomerates, sandstones and shales. It is characterized by coarse, poorly sorted conglomerates and cross-bedded yellow sandstones and shales, which were deposited in a bed-load dominated fluvial system. The detritus was derived from local sources such as the underlying and bordering formations (Fig. 3).

The Río Playas Fm. overlies the upper Cretaceous to Paleocene Casanga, Naranjo and Sacapalca Fms. with an angular unconformity and is partly unconformably overlain by the middle Eocene to lower Oligocene Loma Blanca Fm. (Fig. 5; Hungerbühler, 1997). The stratigraphic relationships suggest an early and/or middle Eocene age for the Río Playas Fm. (Figs. 2 and 5, Appendix A).

The Quingeo Fm. is exposed in two NNE–SSW-trending stripes, one to the east of Cuenca from Quingeo northwards and the other approximately 20 km east of Cañar (Fig. 4). Noblet et al. (1988) and Marocco et al. (1995) mistakenly assigned these outcrops as equivalents of the Biblián Fm., which is a middle Miocene formation exposed in the Cuenca region (see below). The Quingeo Fm. is overthrust along the Santa Ana–Tahual Fault (Fig. 4) by the Maastrichtian (Hughes et al. 1997) Yunguilla Fm. at its western border, and is also covered by late Miocene airborne volcanics (Tarqui Fm.). Steinmann (1997) named the new formation after village of Quingeo and its type locality is at the road along the Río Quingeo (730000/9666800 to 728000/9668000). The outcropping Quingeo Fm. is ≈ 1200 m thick and is certainly only a relic of a much larger basin setting. The formation unconformably overlies partly the Yunguilla Fm. (Figs. 4 and 6) and the thick volcanic Chinchín Fm. However, the latter can only be implied from poor quality outcrops. The Quingeo Fm. displays a succession of 10–30 m thick, repeated fining-upward cycles containing poorly sorted, channelized conglomerates at the base, and sandstones and red and purple siltstones at the top (Fig. 6). Deposition occurred in a low-sinuosity, mixed-load fluvial system (Steinmann, 1997). Flow direction measurements imply that the material was derived from the east, which is also suggested by the presence of metamorphic and quartz vein pebbles from the early Cordillera Real, as well as pebbles from the Yunguilla Fm. Nine ZFT ages on tephra range between 42.2 ± 4.4 and 34.9 ± 4.0 Ma, which

place the Quingeo Fm. in the middle to late Eocene (Fig. 2, Appendix A).

A small outcrop of sediments in the Cordillera Occidental in the lake district west of Cuenca and north of Laguna Luspa (696981/9693135) also yielded a late Eocene ZFT ash age of 37.1 ± 3.8 Ma (Appendix A; Steinmann, 1997). The lithology and age of the sediments suggested that they are part of the Quingeo Fm. (Steinmann, 1997). However, new mapping (Dunkley and Gaibor, 1997b) correlates this outcrop with the volcanic Chulo unit (which also comprises sedimentary intervals) of the Saraguro Group. It appears that the present dating provides an other late Eocene age for the lower part of the Saraguro Group.

2.2.2. Catamayo–Gonzanamá area

With the exception of regional studies by Sauer (1965), Sigal (1969) and Kennerley (1973, 1980), southern Ecuador has received little geological attention. Previous interpretations (Kennerley, 1980) assumed a Paleocene age for the Gonzanamá Fm., because it is bounded both at the top and bottom by the Paleocene Sacapalca Fm. However, regional mapping by Hungerbühler (1997) shows that the lower contact with the Gonzanamá Fm. is unconformable on the Sacapalca Fm. The Sacapalca Fm. occurs again at the top of the Gonzanamá Fm. as a thrust sheet (Fig. 3).

The Gonzanamá Fm. occurs in discontinuous outcrops between the towns of Gonzanamá, Nambacola and Santa Rita (Fig. 3) and overlies the Paleocene to Oligocene volcanic Sacapalca and Loma Blanca Fms. with an angular unconformity. Dominant lithologies include evenly bedded calcareous sandstones, sandy siltstones and minor conglomerates (Fig. 5), and the thickness of the formation varies between 500 and 1000 m. Gypsum veins and sulfur impregnations are abundant. The sandstones show regionally varying thicknesses (10–50 cm) and thick beds in the western outcrops have scoured soles, trough cross-bedding and wave ripple laminations. The conglomerates are rich in volcanic clasts. Thin beds of oolitic calcarenites are intercalated in the series close to the town of Santa Rita. These, and other calcareous sandstones, yielded a rich fauna of ostracods, bivalves and gastropods, and the following ostracods

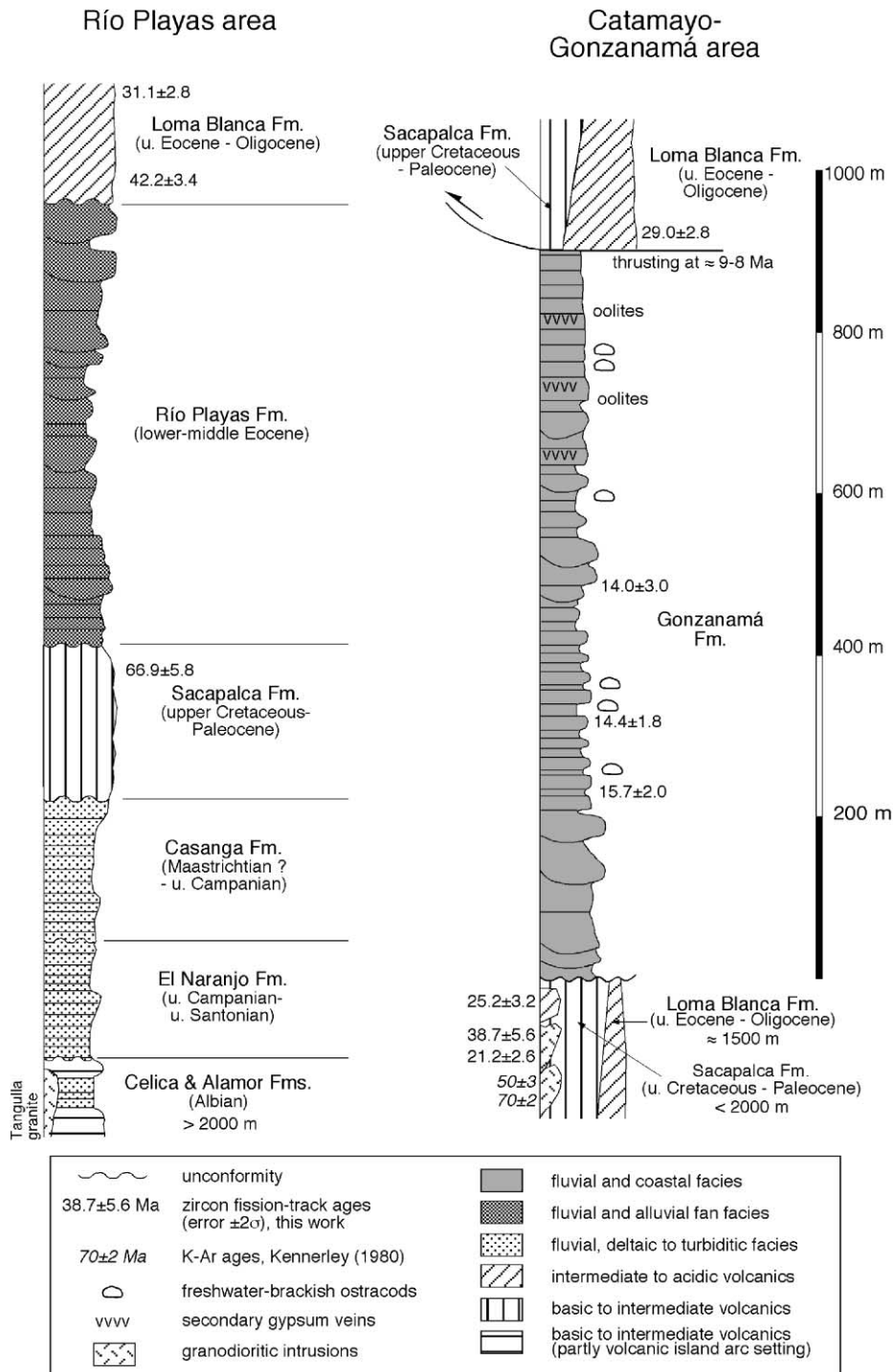


Fig. 5. Composite stratigraphic sections of the middle Miocene series in the Catamayo–Gonzanamá area, and the Eocene Río Playas series at Casanga. From Hungerbühler (1997). Cretaceous stratigraphy after Jaillard et al. (1996, 1999). The Tangulla granite intrusion is possibly of middle Eocene age (E. Jaillard, personal communication, 2001).

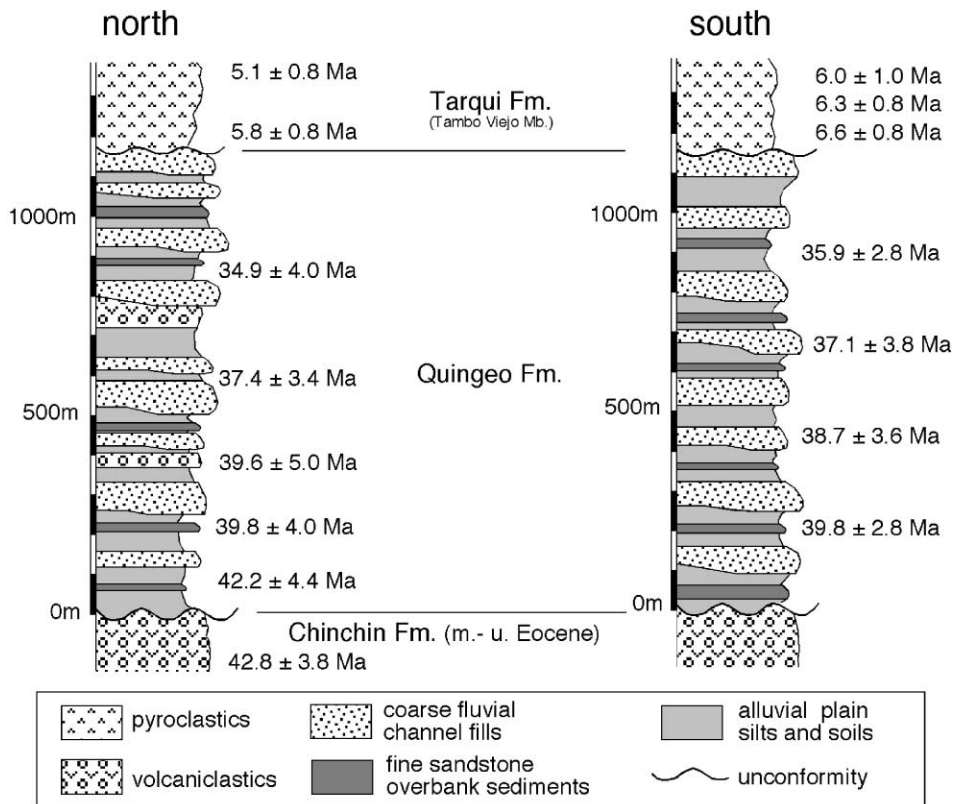


Fig. 6. Composite stratigraphic sections with ZFT ages (2σ errors) of the Quingeo Formation in the Cuenca area. From Steinmann (1997).

have been identified (Peterson et al., 2002): the brackish water species *Cyprideis gonzanamensis* and *Vetustocytheridea splendens*, and several freshwater species consisting of *Heterocypris* sp., *Bradleystrandesia serena*, *Potamocypris santaritaensis*, and other *Cypridid* species. Three acidic tephra in the lower and middle part of the formation gave ZFT ages of 15.7 ± 2.0 , 14.4 ± 1.8 and 14.0 ± 3.0 Ma (Fig. 5, Appendix A; Hungerbühler, 1997) correlating with the middle Miocene. Sedimentary facies and faunal data suggest the formation was deposited in a marginal marine setting with brackish lagoonal and lacustrine environments in the NW and distal fluvial environments in the SW (Hungerbühler 1997).

Jaillard et al. (1996) provide a detailed description of the ≈ 400 -m-thick Catamayo Fm. in the Catamayo area. The lower part predominantly consists of shales, minor sandstones and limestones, with abundant gypsum veins. The middle part is rich in coarse

sandstones and conglomerates and the upper part is once again dominated by shales, with minor sandstone and limestone intercalations. Volcanic clasts prevail in the lower and middle sequences, while in the upper, metamorphic rock fragments are widespread. South of Catamayo village, the Catamayo Fm. rests unconformably on the Sacapalca Fm. and the sediments are overthrust by the volcanic Loma Blanca Fm. (Fig. 3). We were not able to provide chronostratigraphic ages of the formation and poor quality outcrops render it difficult to provide direct correlation with the Gonzanamá Fm. to the south. However, the facies interpretations of Jaillard et al. (1996), suggesting coastal flood and sabkha plain (lower part), fluvial (middle part) and coastal plain environments with marine incursions (upper part) compare closely with the depositional environments found in the Gonzanamá Fm. and in the other middle Miocene series of southern Ecuador.

2.2.3. Malacatos–Vilcabamba area

The sedimentary series in this region partly overlies Paleozoic meta-sedimentary rocks (Chiguinda

unit; Litherland et al., 1994) of the western border of the Cordillera Real and the lower Oligocene volcanics of the Loma Blanca Fm. (Figs. 3 and 7). We

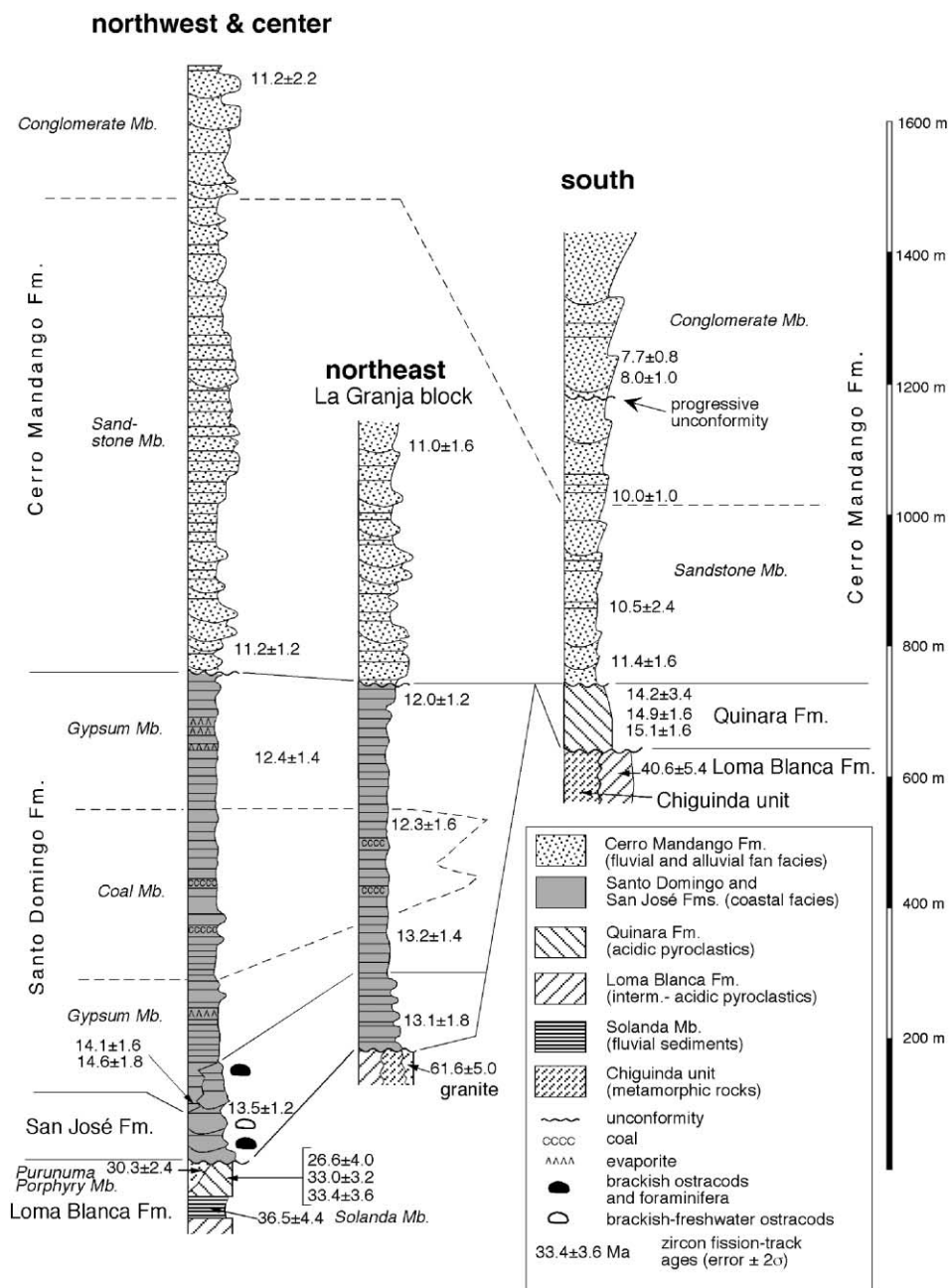


Fig. 7. Composite stratigraphic sections of the middle to late Miocene formations in the Malacatos–Vilcabamba area. From Hungerbühler (1997).

distinguish three tectono-stratigraphic domains located in the south, the northwest and center, and the northeast (La Granja block) (Fig. 7). The latter is separated from the remainder of the basin by a west verging reverse fault (Hungerbühler, 1997). We have applied a modified version of the stratigraphic nomenclature of Kennerley (1973), in which the formation names used here are different from the Loja area.

The Quinara Fm. is mainly exposed in the southern domain and was introduced by Hungerbühler (1997; type locality 694650/9523780) (Fig. 7). It is dominated by white or grey acidic vitric tuffs, ignimbrites and a few lithic (metamorphic grain) tuffs and volcanic breccias and has a maximum thickness of 300 m. The Quinara Fm. unconformably lies on Paleozoic metamorphic units and is both unconformably and conformably overlain by the Cerro Mandango Fm. The pyroclastic lithologies of the Quinara Fm. were previously correlated with the Loma Blanca Fm. (Kennerley and Almeida, 1975a). However, three ZFT age measurements (Hungerbühler, 1997) of 15.1 ± 1.6 , 14.9 ± 1.6 and 14.2 ± 3.4 Ma (Appendix A) place the formation in the early middle Miocene. The Quinara Fm. has no correlative volcanic formation in the region and it is assumed to represent the remnants of a local volcanic event.

The San José Fm. unconformably overlies the Loma Blanca Fm. in the La Granja block and in the northwestern and central part (Fig. 7). In addition, an isolated exposure occurs 2 km W of the village El Tambo (Fig. 3). The lowermost sedimentary series in the area were assigned to the formation by Hungerbühler (1997, type locality in the village of San José, 688650/9537000), and it partly replaces the Algarobillo Fm. of Kennerley (1973). The formation mainly consists of calcareous sandstones of variable thickness and micritic limestones. In addition, there are bioclastic breccias, sheet-like gravel beds with volcanic pebbles, minor bioturbated shales with disk-shaped non-fossiliferous micritic limestone concretions. Amalgamation of coarse sandstone beds is frequent. Horizontal and low angle tabular cross-bedding with abundant reactivation surfaces and hummocky cross-bedding are observed. The maximum thickness of the formation is 200 m. The calcareous sandstones and limestones yielded

abundant ostracods, bivalves, gastropods, scaphopods, foraminifera (*Trochaminita irregularis*, *Quinqueloculina* sp. aff. *seminulum*) charophyte nucleus, fish teeth and bones. The ostracod fauna (Peterson et al., 2002) consists of the recently described freshwater Cypridae (*Cypridopsis tamboensis*, *Bradleystrandesia serena*, *Candona harrisi*, several unnamed *Cyprid* spp.) and the brackish water Cytheridae (*Cyprideis malacatensis*, *Cyprideis* sp. aff. *schedogymnos* Muñoz-Torres, Whatley and van Harten, 1998, *Vetustocytheridea splendens*). Two tephra in the formation yielded ZFT ages of 13.5 ± 1.2 Ma and 13.1 ± 1.9 Ma-late middle Miocene (Fig. 7, Appendix A). The marine foraminifera and brackish and freshwater ostracod fauna imply a coastal marine depositional environment with varying freshwater input. The sandstone facies association compares partly with sand waves formed by tidal currents in estuarine channels and partly with channel mouth bars at a delta front. The thin calcareous sandstones and limestones represent flood plain and lagoonal deposits. In conclusion, the faunal and facies information suggests that the San José Fm. was deposited in a tidal dominated delta environment.

The Santo Domingo Fm. was described by Hungerbühler (1997; type locality Quebrada Santo Domingo, 690000/9537750) and replaces the upper part of the Algarobillo Fm. and the Cabalera Fm. of Kennerley (1973). The formation, which occurs in the northern domain and in the La Granja block, is divided into two interfingering members (Fig. 7). The thickness is variable up to a maximum of 700 m. The *Coal Mb.* is composed of repeated sequences of well-bedded bituminous black shales, siltstones and minor, partly channelized sandstone intervals with cross-stratification and wave ripple laminations. The bituminous shales have a hydrocarbon yield of 60–80 kg/tonne (Litherland et al., 1994). Calcareous concretions and clastic dykes are ubiquitous and several laterally continuous coal seams reach a thickness of 2 m (Putzer, 1968). The *Gypsum Mb.* consists of greenish, medium-grained, well-sorted sandstones with wave and vortex ripple laminations and laminated siltstones. The gypsum occurs in the sandstones both as primary prismatic vertical needles and flat lying reworked crystals in funnel shaped breccia lenses. Secondary gypsum is also very abundant as

up to 20 cm thick. The gypsum is concentrated also in large-scale fold saddle structures, where it is currently mined near La Merced and in the Quebrada Santo Domingo. Moderately and poorly preserved gastropods (similar to the *Dyris* species in the Loja area), shrimps (similar to *Penaeus maddenii* in the Loyola Fm. in the Cuenca area), ostracods and fish teeth were found, all of which could not be determined in more detail. Six tephra yielded ZFT ages, ranging between 14.6 ± 1.8 Ma and 12.0 ± 1.2 Ma (Fig. 7, Appendix A). The older ages overlap with the ages obtained in the San José Fm. confirming an interfingering of the two formations, as is also suggested from field mapping. The facies and mineralogy of the Santo Domingo Fm. imply a coastal deltaic depositional setting with predominant tidal flat and backswamp environments, similar to the San José Fm. This is also corroborated by the occurrence of primary gypsum and coal seams in N–S trending exposures parallel to the reconstructed shoreline (Hungerbühler, 1997). The high sulfur content of the coals (Putzer, 1968) is suggestive of formation in a coastal environment, such as saline marshes (Styan and Bustin, 1984). Furthermore, the alternation of gypsum and coal bearing intervals (containing fossil leaves indicating a low altitude tropical climate; e.g. Berry, 1945) suggests that deposition occurred under varying dry and humid conditions along the edge of a coastal plain (Hungerbühler, 1997).

The main outcrops of the Cerro Mandango Fm. (Kennerley, 1973) occur within a SSE–NNW oriented syncline between Vilcabamba and Malacatos (southern and central tectono-stratigraphic domain, where it is up to 1000 m thick), and in the La Granja block (Figs. 3 and 7). It overlies the Santo Domingo Fm., and older metamorphic rocks with angular unconformities and rests partly conformable on the Quinara Fm. The observed succession of a lower Sandstone Mb. and an upper Conglomerate Mb. (Hungerbühler, 1997) reflects the general coarsening-upward trend of the formation (Fig. 7). The upper member is thickest in the southern part of the basin where spectacular internal progressive unconformities are present (Hungerbühler, 1997). The Sandstone Mb. consists of cross- and tabular-bedded, medium- and coarse-grained sandstones, sheet-like gravel beds with metamorphic pebbles, and minor siltstones. The Sandstone Mb. grades up into the

Conglomerate Mb., which is characterized by broadly channelized, clast (metamorphics) supported conglomerates, coarse sandstones displaying sheetflow features and a few siltstones. In places, white crystal and lithic (metamorphic clasts) tuffs, tuffaceous sandstones and pumice breccias are intercalated providing suitable zircon bearing marker beds for ZFT analysis. Eight samples in the formation yielded ages ranging between 11.4 ± 1.6 and 7.7 ± 0.8 Ma (Fig. 7, Appendix A). The transition from the Sandstone Mb. to the Conglomerate Mb. occurred at ≈ 10 Ma and the progressive unconformities in the Conglomerate Mb. are dated at ≈ 8 Ma (Fig. 7). The facies succession depicts the transition from a bed-load dominated sandy fluvial system to an alluvial fan system, which prograded from the south to the north. The source of the reworked material was exclusively situated in metamorphic rocks of the Cordillera Real (Hungerbühler, 1997).

2.2.4. Loja area

A stratigraphic scheme for the Loja area was initially established by Kennerley (1973). Later, a nomenclature that was consistent with the Malacatos–Vilcabamba area was applied (Kennerley and Almeida, 1975a,b). However, Hungerbühler (1997) showed that the lithofacies developed differently in the two areas and different ages are obtained from similar facies. Therefore, we chose to mainly follow the stratigraphic nomenclature of Alvarado (1967), and partly that of Kennerley (1973) in the Loja area. In addition, the sedimentary series in the Loja area are tectonically divided by a major $\approx$ W verging post-depositional reverse fault into western and eastern parts, juxtaposing different sediment sequences at their mutual boundary (Figs. 3 and 8). The sediment series lie unconformably on older metamorphic rocks (Chiguinda and Agoyán units; Litherland et al., 1994) on both sides of the reverse fault. First the western, then the eastern part will be described (Fig. 8).

The Trigal Fm. consists of coarse sandstones, fine conglomeratic sheet layers and minor siltstones, which were deposited in bed-load dominated fluvial systems and by sheet floods. In contrast to the other formations in the Loja area, which contain abundant metamorphic clasts, the Trigal Fm. mainly carries

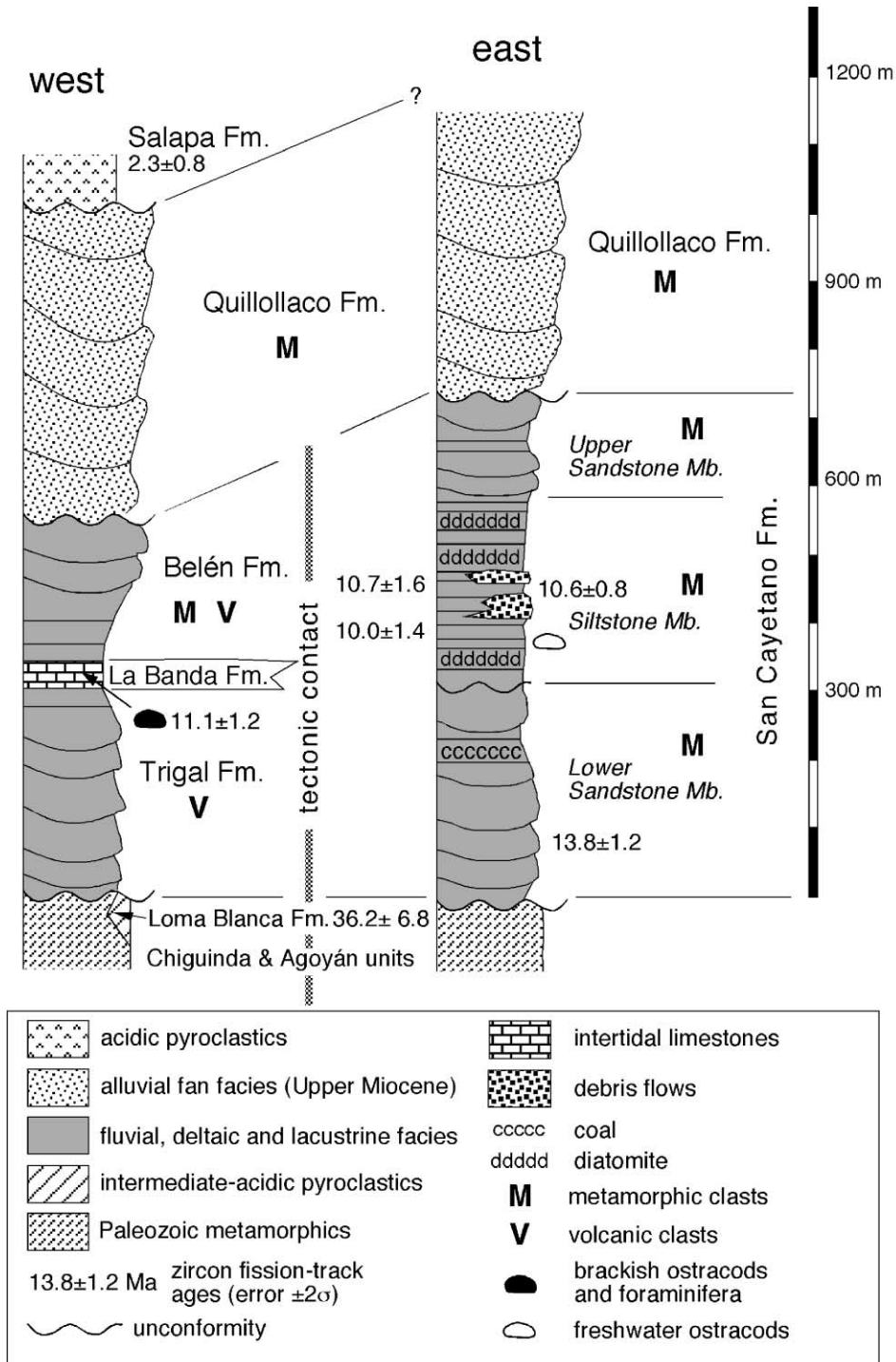


Fig. 8. Composite stratigraphic sections of the middle to late Miocene formations in the Loja area. From Hungerbühler (1997).

volcanic clasts. The formation reaches a thickness of ≈ 150 m but could not be directly dated. However, the conformably overlying La Banda Fm. yielded a ZFT age of 11.1 ± 1.2 Ma (Hungerbühler, 1997), suggesting that the Trigal Fm. is most likely a middle Miocene sequence. The La Banda Fm. has a maximum thickness of only 10–20 m, but it is very distinct and laterally continuous (Fig. 8). It consists of: (1) thick-bedded white limestones; (2) rhythmic thin-bedded, finely laminated limestones and marly shales; (3) thin chert beds; (4) intraformational limestone breccias; and (5) fine-grained yellow sandstones. Secondary gypsum is ubiquitous in fractures and seams and the presence of primary gypsum can be inferred from the occurrence of pseudomorphic calcite. Sheet-like algal mat laminations, dessiccation cracks, wrinkle marks and bioturbations are observed. An abundant ostracod fauna was determined (Peterson et al., 2002) and includes the brackish water *Cyprideis lojaensis* and *Cyprideis malacatensis*, the freshwater *Lymnocythere* sp. and *Bradleystrandesia serena*, and an unspecified *Cypridid* sp. In addition, the foraminifera *Trochaminita irregularis* and *Discorbis* sp. were found. The sedimentary facies association and biogenic content shows characteristic features of a lagoonal environment with supra- to intertidal deposits where repeated drying cycles occurred. The increasing frequency of sandstone layers marks the upward transition from the La Banda Fm. into the Belén Fm. (Fig. 8). The 300-m-thick Belén Fm. is characterized by coarse channel sandstones and conglomeratic lag deposits, which alternate with well-bedded finer, large-scale cross-bedded sandstones and minor shale beds. Metre-scale thick slumps are observed in the lower part of the formation. The observed facies variation is interpreted to indicate a transition from a lagoonal (La Banda Fm.) to a deltaic and mixed-load fluvial environment (Belén Fm.).

The coeval San Cayetano Fm. in the eastern Loja area (Fig. 8) is divided into three members separated by transitional boundaries (Hungerbühler, 1997). Due to strong tectonic deformation and transitional sedimentary contacts, the thicknesses of the formation and of the members are difficult to assess. The San Cayetano Fm. possibly achieves a total thickness of 800 m. The *Lower Sandstone Mb.* consists of thick sandstones, channelized conglomerates, minor shales

and several coal seams. A fining-upward trend towards the *Siltstone Mb.* is observed, which in turn is characterized by laminated brown, grey and white shales (partly silicified), abundant diatomite layers, and a few pyroclastic horizons. In addition, two distinct 3–5-m-thick breccia layers were observed (Fig. 8), indicating that catastrophic debris flows entered the otherwise rather quiet depositional realm. The *Upper Sandstone Mb.* displays a similar lithologic character as the lower one, but is generally finer grained and shows a coarsening-upward trend. The Lower Sandstone and the Siltstone Mbs. of the San Cayetano Fm. are rich in wood and leaf remains, fish skeletons, gastropods and diatoms. The macroflora (leaves) were studied by Berry (1945), who suggested that sedimentation occurred in a tropical climate at low altitude. Hungerbühler (1997) combined these earlier paleontologic results with the observed sedimentary facies and concluded that the Siltstone Mb. was deposited in a clastic dominated freshwater lake situated in a tropical environment at low elevation. The lake formed during the time between the retreat of the mixed-load fluvial system (Lower Sandstone Mb.) and the subsequent progradation of the Upper Sandstone Mb., which filled the lake. Four ZFT ages from the Sandstone Mb. (13.8 ± 1.2 Ma) and the Siltstone Mb. (10.7 ± 1.6 to 10.0 ± 1.4 Ma) (Fig. 8, Appendix A) are available.

The Quillollaco Fm. is present in the eastern and western Loja area and overlies the older formations with an angular unconformity. The formation reaches maximum thickness of 600 m east of the town of Loja and generally consists of tightly stacked metamorphic clast-supported conglomerates and lens-shaped sandstones. Very coarse alluvial fan facies and several matrix-supported breccia intercalations occur to the west of the town of Loja. The clast imbrications imply that the transport direction was E to W. A transition to a braided river system has been identified in the central part, and a general coarsening-upward trend and progradation from the east towards the center is observed. Therefore, it is inferred that the alluvial fan of the Quillollaco Fm. prograded westwards due to steepening of the source terrane in the Cordillera Real. The high-energy depositional environment did not allow any datable pyroclastic beds to be preserved. However, lithologic similarities with the Cerro Mandango Fm. and a

similar deformation history, suggest the Quillollaco Fm. may be a late Miocene sequence.

Volcanic formations are rare in the Loja area and Kennerley (1973) grouped them under the name Salapa Fm. Furthermore, he considered them to lie beneath the sediment series. However, field relationships and a single ZFT age of pyroclastic strata exposed north of Loja (2.4 ± 0.8 Ma, Fig. 8 and Appendix A) show that the Salapa Fm. is in fact younger than the sedimentary series. These lithic clast and glass rich tuffs (strongly kaolinitized) are preserved in small scattered outcrops and represent the youngest pyroclastic rocks ever analyzed in the southern Sierra of Ecuador (Hungerbühler, 1997).

2.2.5. Girón–Santa Isabel area

Restricted access to the region around the villages of Girón and Santa Isabel prevented a detailed study of the sedimentary series until the 1960s. Sauer (1965), Bristow (1973), and Randel and Lozada (1974) correlated the sedimentary rocks with those in the Cuenca area. More recently, Hungerbühler (1997) and Steinmann (1997) found that continental sediments prevail in the Girón–Santa Isabel area whereas coeval coastal deltaic facies are present in the Cuenca area.

A small N–S-striking belt of ≈ 100 -m-thick sedimentary rocks occurs between the volcanic Saraguro and Santa Isabel Fms in the southern part of the Santa Isabel area. The Jacapa Fm. (Fig. 9, type locality in the Quebrada Cajamarca 683350/9621350) of Hungerbühler (1997) and Hammer (1998) consists of fine to coarse sandstones, massive siltstones and thin limestone beds, which are in unconformable contact with the bounding volcanic formations. The Jacapa Fm. was deposited during the initial stages of basin formation (proto-Santa Isabel Basin in Hungerbühler, 1997) and contains a rich freshwater ostracod fauna (Peterson et al., 2002) with several Cyprididae and Limnocytheridae species, including the new species *Cypridopsis isabelensis*, *Candona harrisi* and *Cytheridella purpuri*. Sedimentary facies and faunal content suggest that deposition occurred in lacustrine and fluvial plain environments. The Jacapa Fm. is bracketed between the underlying Saraguro Fm. and overlying Santa Isabel Fm. with ZFT ages of 19.1 ± 1.4 and $18.4 \pm$

2.0 Ma, respectively (Fig. 9). Consequently, a general early Miocene age can be inferred.

The Burrohuayacu Fm. was described by Hungerbühler (1997) and Helg (1997). In the type region it occurs between the village of Uchucay (681000/9630000) and the Quebrada Burrohuayacu (684000/9632000). The new formation replaces in the area the Ayancay Group, which was previously defined by Kennerley et al. (1973) and Randel and Lozada (1974). The Burrohuayacu Fm. displays strong lateral and vertical variations. Along the western margin its thickness is in the range of a few tens of meters and it increases to ≈ 800 m in the south/southeast (see also Figs. 14 and 15). The Burrohuayacu Fm. unconformably overlies the Saraguro Fm. and lower parts of the Santa Isabel Fm. and is in turn subdivided into a *Volcaniclastic*, a main *Sandstone* and in a *Conglomerate Mb.* (Fig. 9). The Volcaniclastic and Conglomerate Mbs. are of local importance in the area. The thin Volcaniclastic Mb. consists of well bedded volcaniclastic sandstones and conglomerates (Saraguro Group clasts), which were deposited in a low-energy, fluvial environment. The overlying Sandstone Mb. is a thick series of red siltstones with laterally and vertically stacked, decametric channelized sandstone bodies, which suggest that deposition occurred in a sand-dominated braided river system. Paleocurrent measurements (channel and foreset orientations) reveal bimodal NE and SE directions, indicating that transport mainly occurred towards the NE and lateral input (Conglomerate Mb.) came from the NW (Helg 1997). The Conglomerate Mb. is composed of coarse, clast-supported, poorly sorted conglomerates (clasts from the Saraguro Gp.), which grade towards the SE into the Sandstone Mb. Laterally continuous sheet-like sandstone beds are observed in the transition zone. The Conglomerate Mb. interfingers with the Sandstone Mb. and represents a local alluvial fan that entered the basin from the NW. Ten homogeneous ZFT ages from the Burrohuayacu Fm. range between 14.7 ± 1.2 and 10.5 ± 1.6 Ma (Fig. 9, Appendix A).

According to the facies and age relationships, the Girón Fm. in the Girón area represents a different and younger formation than the Burrohuayacu Fm. (Fig. 9). Hungerbühler (1997) defined the Girón Fm. at the type locality 703900/9646500. The formation

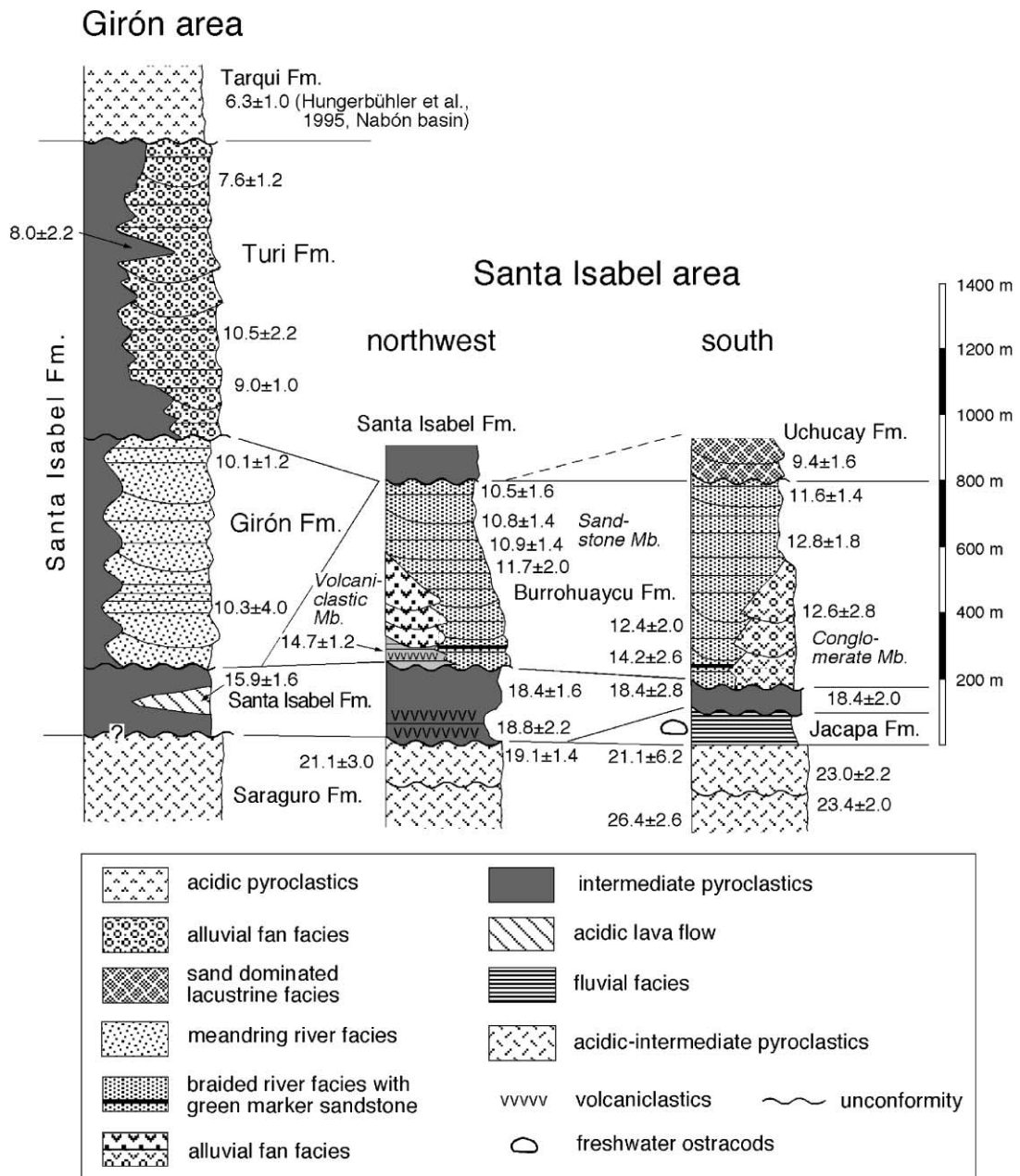


Fig. 9. Composite stratigraphic sections with ZFT ages (2σ errors) of the middle to late Miocene formations in the Girón–Santa Isabel area. From Hungerbühler (1997).

is up to 800 m thick, unconformably overlying the lower parts of the Santa Isabel Fm. A physical contact with the Burrohuaycu Fm. was not observed. The Girón Fm. is dominated by red and grey channelized, medium to coarse grained sandstones with

intercalations of fine sandstones, silts and laminated shales. A mixed-load (meandering) fluvial system is inferred, which drained from S to N (Hungerbühler, 1997). Two ZFT ages of 10.3 ± 4.0 and 10.1 ± 1.2 Ma have been recorded.

The Uchucay Fm. overlies the Burrohuayacu Fm. with a marked angular unconformity in the Santa Isabel area. This unit was also recognized by Kennerley et al. (1973) (Figs. 9 and 15). The formation has a maximum thickness of 100 m and mainly consists of white, laminated siltstones intercalated with poorly sorted conglomerates (clasts are from the Saraguro Group) and sandstones. The abundance of laminated siltstones suggests that deposition occurred in a lake, which had a high detrital input (Hungerbühler, 1997; Hammer, 1998). According to heavy mineral analyses (Helg, 1997), increasing amounts of epidote, garnet, muscovite and actinolite in the upper part of the Burrohuayacu Fm. (younger than ≈ 11 Ma) and in the Uchucay Fm. indicate that in the supplying Cordillera Real in the SE erosional incision involved medium-grade metamorphic rocks. A reworked tephra layer in the Uchucay Fm. yielded a range of Miocene ZFT grain ages. However, the youngest population that is assumed to represent the eruption age has an age of 9.4 ± 1.6 Ma.

The Girón Fm. is overlain with an angular unconformity by the 1000-m-thick Turi Fm. in the Girón area (Fig. 9), i.e. it is much thicker here than in the Cuenca area (see below). Randel and Lozada (1974) already mapped coarse clastic deposits located to the NW of the Girón village as Turi Fm. New roadcuts and ZFT analyses of rocks exposed over a larger area surrounding San Fernando (Fig. 3) are clearly correlated with the Turi Fm., instead of the previously mapped Tarqui Fm. The Turi Fm. consists of tuffaceous coarse sandstones, channelized, volcanic clast-supported conglomerates, matrix-supported volcanic breccias and minor tuff layers. A general coarsening-upward trend is observed. Deposition is inferred to have occurred in alluvial fan and proximal bed-load dominated fluvial systems. Vertical and lateral grain size trends suggest that the systems prograded towards the E and SE under an increasing topographic gradient. Four tephra provided ZFT ages ranging between 10.5 ± 2.2 and 7.4 ± 1.2 Ma (Appendix A). The interfingering volcanic Santa Isabel Fm. yielded a ZFT age of 8.0 ± 2.2 Ma (Fig. 9).

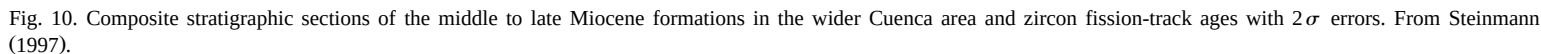
2.2.6. Cuenca area

The largest outcrops of Miocene sediments in Ecuador are observed in the Cuenca region (Fig. 1). The formations occur in a NNE–SSW extending

exposure, which covers a geographic surface of about 3000 km² (Fig. 4). Bristow (1973), Noblet et al. (1988) and Marocco et al. (1995) established a stratigraphic scheme. However, several new members are introduced in the present study, and the chronostratigraphic correlations are revised. The sediment series has a total thickness of 2400–3500 m and can be divided into two main sequences, which are separated by a regional unconformity (Fig. 11).

The lower sequence consists of fluvial, deltaic and brackish delta plain deposits in which metamorphic pebbles from the Cordillera Real are ubiquitous. Traversing from the bottom to the top of the sequence, it can be divided into the Biblián, Loyola, Azogues and Mangán Formations. The Biblián Fm. unconformably overlies the Saraguro Fm. and the intervening hiatus amounts ≈ 7 –10 Ma (Fig. 10). The sediments consist of alternating purple, red and green claystones with frequently erosive, decimeter to meter scale fine to medium grained sandstone layers. Good outcrops are rare, although the formation was well exposed during construction of the highway from Cuenca to Azogues (during 1995/1996). Deposition occurred on the flood plains of a suspension-load dominated fluvial system although meander channels are rarely exposed. Two ash layers provided ZFT ages of 14.7 ± 2.4 and 12.3 ± 1.6 Ma (Appendix A). The Eocene Quingeo Fm. (see above) was previously correlated with the Biblián Fm., which was assumed to have a total thickness of 1500 m (Noblet et al., 1988). However, the Biblián Fm. as defined here only has a maximum thickness of 300 m.

The ca. 450-m-thick Loyola Fm. generally conformably overlies the Biblián Fm. (Fig. 10), although low angle unconformities are observed in several places (e.g. Noblet et al., 1988; Marocco et al., 1995). While it generally weathers to an orange colour in most outcrops, fresh roadcuts reveal black, finely laminated, pyrite-rich shales with fine (northern and central part) to coarse (south) sandstone intercalations. The sandstone layers represent tongues of the Azogues Mb., which interfinger with the Loyola Fm (Fig. 11). The Loyola Fm. is well known for its rich fauna of molluscs, gastropods, fish skeletons, shrimps, marine crab claws and ostracods (Bristow and Parodiz, 1982; Nuttall, 1990; Feldmann et al., 1993; Steinmann, 1997). In addition, Peterson



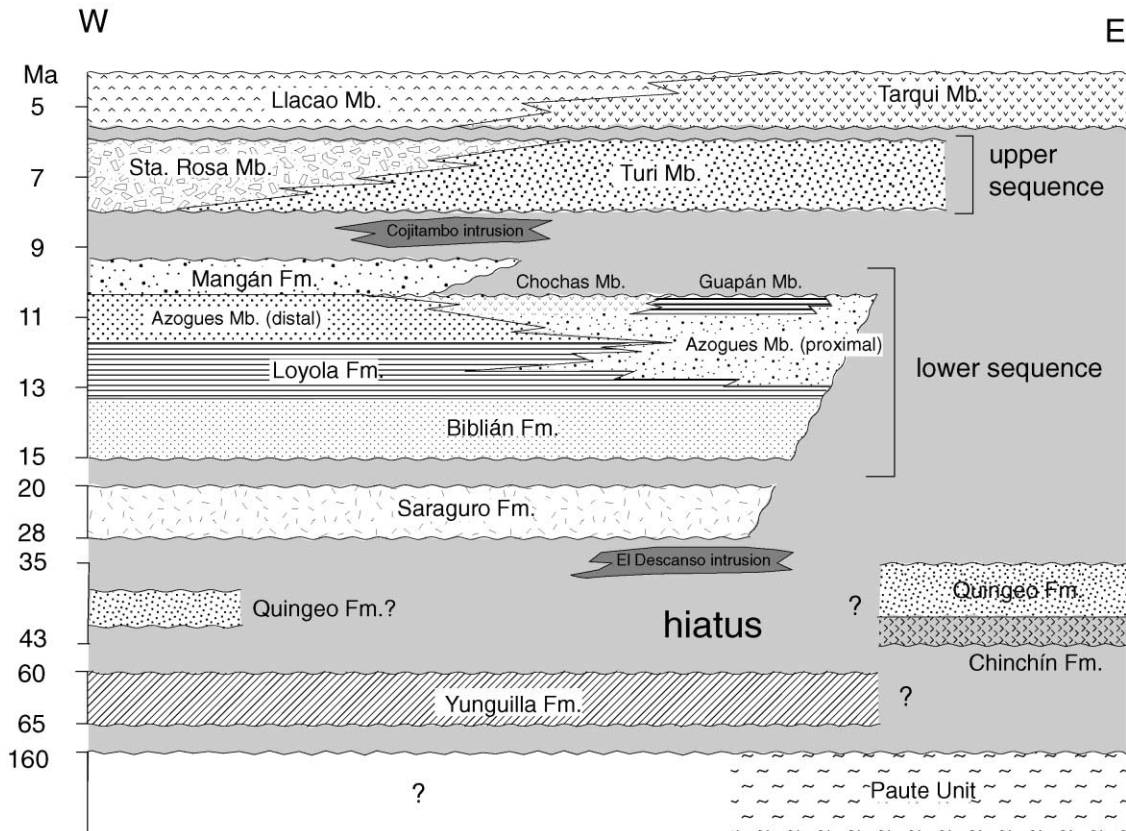


Fig. 11. Time–space diagram of the formations in the Cuenca area in an E–W profile. The Miocene sedimentary series are arranged in two sequences. The lower one is the Pacific Coastal stage sequence, the upper one is the Intermontane stage sequence. Modified from Steinmann (1997).

et al. (2002) identified abundant brackish water ostracods *Vetustocytheridea bristowi* and the new species *Cyprideis malacatensis*, as well as the marine ostracod *Macrocypris* sp. The Loyola Fm. has a complicated regional facies pattern of marine deltaic, distal fluvial and locally lacustrine facies deposited in low energy environments. Prodelta deposits prevail in the northern and northwestern parts, and sandy delta front and fluvial deposits in the E and SE. ZFT ages from five tephra range between 13.9 ± 1.4 and 11.1 ± 1.0 Ma (Fig. 10, Appendix A).

The Azogues Fm. is divided into three members (Figs. 10, 11). The *Azogues Mb.* (Steinmann, 1997) is 450–500 m thick and comprises more than the lower two thirds of the formation. The freshest outcrops are exposed along the new road from Cuenca to Azogues. The contact with the underlying Loyola

Fm. is gradual although interfingering is observed in several locations. The member consists of grey, coarse grained, massive sandstones and siltstones and minor shales with a regional fining trend from the S to NW. Massive, partly conglomeratic channel fill deposits with trough cross-bedding and coarsening-upward trends are observed in the proximal part in the S. They are intercalated with siltstones, thin sandstones, paleosols and thin coal layers. Silicified wood fragments and leaf prints are frequent. Towards the NW, in the distal facies, massive structureless sandstones dominate over brown, partly laminated siltstones and shales. The sandstones were deposited from mass flows and show some bedding features characteristic of turbidites (Noblet et al. 1988; Noblet and Marocco, 1989). High sedimentation rates of ≈ 1 mm/year (compacted) gave rise to

depositional instabilities, which are evident from the presence of spectacular syn-sedimentary soft sediment deformation features such as slumps and diapirs exposed on a decametric scale (Steinmann, 1997). The facies of the Azogues Mb. in the SW records deposition in a bed-load and mixed-load fluvial and alluvial plain system, which transformed into a fluvially dominated delta system (*sensu* Galloway and Hobday, 1996) towards the NW. Both systems prograded towards the NW. Five ZFT age determinations from the Azogues Mb. range between 13.0 ± 1.0 and 11.9 ± 1.2 Ma (Fig. 10, Appendix A).

The remaining two members of the Azogues Fm. are only locally important (Figs. 10, 11). The *Guapán Mb.* (Steinmann, 1997) (≈ 200 m thick), exposed in the syncline structure around and south of Azogues town (Fig. 4), consists of thinly laminated dark brown and black shales with tuffaceous sandstones, white diatomite and clay layers. Plant remains and entire leaves are abundant although no ostracods were found. The facies association suggests that deposition occurred in a large lake, which was probably dammed for some time by the prograding fluvial dominated delta of the Azogues Mb. (Fig. 11). One tephra yielded an age of 11.5 ± 1.4 Ma. The *Cochas Mb.* rarely exceeds a thickness of 100 m and is exposed in scattered outcrops E of Cuenca (Figs. 4 and 10). Steinmann (1997, type locality 734500/9684000) noted that the member is mainly composed of primary volcanic deposits including grey to brown tuff layers, pumice and crystal tuffs, lapilli beds and fine conglomerates. Aquatic reworking on an alluvial plain is evident, which produced coarsening-upward grading of pumice pebbles in individual beds. The volcanic character of the member favoured ZFT analyses and four ages fall within a narrow range between 11.0 ± 1.0 and 10.2 ± 1.2 Ma (Fig. 10, Appendix A).

The Mangán Fm. has a maximum thickness of 1000–1200 m and is exposed in the west of the Cuenca–Ingapirca area. Fresh outcrops are observed between the villages of Ayancay and Deleg (Fig. 4). The Mangán Fm. overlies the Azogues Fm. with an angular unconformity in the central part although it overlies the Loyola Fm. in the north due to the absence of the Azogues Fm. (Fig. 10). The Mangán Fm. displays a coarsening-upward trend with abun-

dant volcanic ash layers. The lower part of the formation consists of alternating grey, green and red shales with sandstone beds depicting climbing ripple stratification and thicker beds with planar and trough cross-bedding. Thicker and coarse, channelized sandstone and conglomerate beds arranged in fining-upward sub-cycles dominate in the upper part. In addition to many thin coal seams, in the upper formation there are also two 3-m-thick coal horizons known (named Washington and Cañari), which have been exploited to the west of Cuenca. The Mangán Fm. was deposited on a fluvially dominated delta plain containing ponds and backswamps (Steinmann, 1997). The coal contains a high amount of sulfur (up to 6 wt.%; O'Rourke, 1978) and terrigenous matter, which is typically found in peats, which developed in saline marches in coastal plain settings (Styan and Bustin, 1984). O'Rourke (1978) suggested that the delta grew at or near sea level. The rich flora and fauna and in particular the pollen discovered in the coals seams led Putzer (1968) to interpret a tropical depositional environment. The presence of the ostracod *Vetustocytheridea bristowi* is significant (Van den Bold, 1976; Peterson et al., 2002) and indicates deposition in brackish water. The gastropods belong to the family of *Neritidae* (*Neritina roxoi*, de Greve, 1938, F. Wesselingh, personal communication, 2000), which also include few brackish and marine forms. The gastropods were not classified to a species level (Nutall, 1990) and therefore, their environmental significance is not constrained. However, the remaining observations suggest that the lower part of the formation was deposited in a coastal delta system (Steinmann, 1997). The mixed-load fluvial character of the facies in the upper formation suggests that the delta system prograded westward. Concordant ZFT ages from four ash layers (9.9 ± 1.2 to 9.5 ± 1.0 Ma, Appendix A) combined with the high formation thickness imply that sedimentation occurred at a very high rate.

The upper sedimentary sequence in the Cuenca area is represented by the volcanic clast-bearing Turi Fm. (Figs. 10 and 11), which is divided into the coeval *Turi* and *Santa Rosa Mbs.* These members overlie the Mangán Fm. with a regional angular unconformity of up to 60° . The ≈ 300 -m-thick Turi Mb. (type locality 721200/9676800 near the village of Turi; Fig. 4) consists of poorly sorted, coarse

conglomerates and cross-bedded sandstones that were deposited in a proximal braided river system. Five samples yielded ZFT ages between 9.6 ± 1.8 and 8.0 ± 1.2 Ma (Fig. 10, Appendix A). The ≈ 150 -m-thick Santa Rosa Mb. was defined by Bristow (1973) although some confusion has arisen in the past regarding type sections. Steinmann (1997) described the most typical outcrops in the area to the south of the village Nazón (733000/9701500). The Santa Rosa Mb. is composed of poorly sorted conglomerates and lenticular coarse sandstones, which were deposited on an alluvial fan that was situated to the west of the depocenter of the Turi Mb. Lateral interfingering of the two members, their geographic distribution (Fig. 4), flow direction indicators and the abundance of volcanic pebbles suggest that they are

the first sequences at their latitude that were sourced from volcanic material of the incipient Cordillera Occidental during the Neogene (Steinmann, 1997). Both members are unconformably overlain by the volcanic Tarqui Fm. (see above).

2.2.7. Nabón area

A generalized stratigraphic nomenclature for the Nabón area was first compiled by Bristow (1976) and later refined by Winkler et al. (1993), Baudino et al. (1994), Hungerbühler et al. (1995) and Winkler et al. (1996). The small Nabón Basin formed and filled during a relatively short time period between ≈ 8.5 and 7.9 Ma (Fig. 12, Appendix A) and the recognition of the palaeomagnetic chron “4r” (late Miocene) in the sediments confirms the ZFT chronostrati-

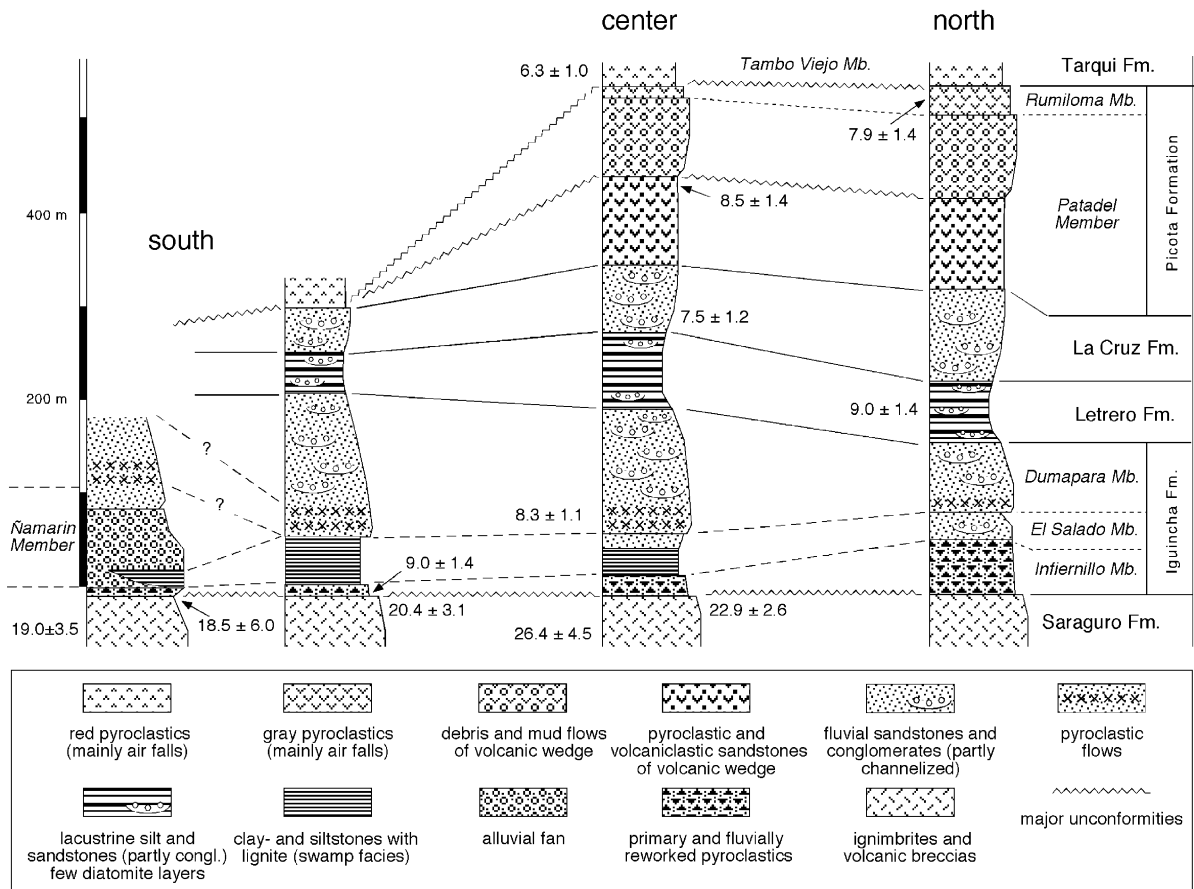


Fig. 12. Composite stratigraphic sections in the Nabón Basin and zircon fission-track ages with 2σ errors. From Hungerbühler et al. (1995).

graphic correlation of Hungerbühler et al. (1995). The total basin sequence is 500–600 m thick unconformably overlying the Saraguro Fm. (part of the Saraguro Group) for which ages of 26.4 ± 4.5 to 19.0 ± 3.5 Ma were obtained (Fig. 12, Appendix A). More details of the chronostratigraphic correlations and lithofacies descriptions are presented in Hungerbühler et al. (1995) and Winkler et al. (1996).

Sedimentation in the Nabón area occurred during a period of varying volcanic activity along the eastern and northern margin of the basin and several syn-eruptive and inter-eruptive stages can be differentiated. The *Iguincha Fm.* is divided into four members (Fig. 12). The first period of continuous sedimentation commenced during an eruptive phase giving rise to mainly ash and pumice beds, which covered the basin floor (*Infiernillo Mb.*). However, reworking of this volcanic material by small rivers and gravity-driven processes is evident. Small alluvial fan systems prograded into the basin from the N and SE (*Ñamarin Mb.*). The overlying and partly interfingering *El Salado Mb.* represents a bed to mixed-load fluvial system with braided channels, which drained the basin from the NE towards the SW. Swamps and peats prevailed in the latter topographic lows (Fig. 12). Pyroclastic flows and falls that form the base of the *Dumapara Mb.* document a subsequent, but short-lived syn-eruptive basin-fill episode. These were overlain by sediments deposited from bed-load dominated river systems, which entered from the NE and E and converged in the lower part of the basin to the south. The overlying clastic lake deposits (with few diatomite layers) of the *Letrero Fm.* (Fig. 12) probably indicate a period of decreased tectonic and volcanic activity. However, abundant detrital input drove the rapid fill of the lake by the meandering fluvial systems of the *La Cruz Fm.* The upper most *Picota Fm.* (Fig. 12) formed a volcanoclastic mass-flow wedge during a further syn-eruptive stage. Subsequently, the basin-fill was partly eroded and incised. The volcanic ashes of the *Tambo Viejo Mb.* (6.3 ± 1.0 Ma) sealed the resulting topography.

The short lifespan of the Nabón Basin was characterized by syn-sedimentary tectonic deformation, which is documented by progressive unconformities along the western margin of the basin, as well as growth folds and faults (Hungerbühler et al., 1995;

Baudino et al. 1994). Maximum shortening occurred in WNW–ESE direction, i.e. perpendicular to the longer basin axis.

3. Late Miocene unconformity

The facies, faunal and chronostratigraphic relationships in the Miocene Inter-Andean domain in southern Ecuador reveal the existence of two distinct sequences, which span the middle to early late Miocene and the late Miocene. They are separated by a major unconformity dated at about 10–9 Ma (Fig. 13). The complete stratigraphic range of both sequences is exposed in the Cuenca, Girón–Santa Isabel, Loja and Malacatos–Vilcabamba areas. However, only the younger sequence is found in the Nabón area. The older sequence is recognized in the Catamayo–Gonzanamá area. With the exception of the Girón–Santa Isabel area, the lower sequence hosts marginal marine facies elements (see Fig. 13): tidal flat and lagoon in the Catamayo and Gonzanamá Fms.; tidal flat, backswamp and delta plain in the San José and Santo Domingo Fms.; supratidal/intertidal lagoon and delta plain in the La Banda, Trisal and Belén Fms.; prodelta, delta front and delta plain in the Loyola, Azogues and Mangán Fms. In the upper part of the lower sequence, increased sediment supply and generally west-directed delta progradation is common in these fluvial deltaic facies (e.g. lower part of Cerro Mandango Fm., Mangán Fm.). Important freshwater lake deposition that was most likely driven through damming up by the deltas, is recorded in the Siltstone Mb. of the San Cayetano Fm. and in the Guapán Mb. of the Azogues Fm. The Burrohuaycu and Girón Fms. were continuously deposited in continental fluvial depositional environments in the Girón–Santa Isabel area, and they were unconformably overlain by late Miocene continental formations (Uchucay, Turi Fms.) (Fig. 13). The mingling of brackish and freshwater ostracods is observed in the various delta and coastal plain deposits, where the percentages of brackish and freshwater forms often approach 50% (Peterson et al., 2002). The ostracod assemblages suggest that an euryhaline depositional setting persisted, which agrees with the interpreted deltaic and coastal plain depositional en-

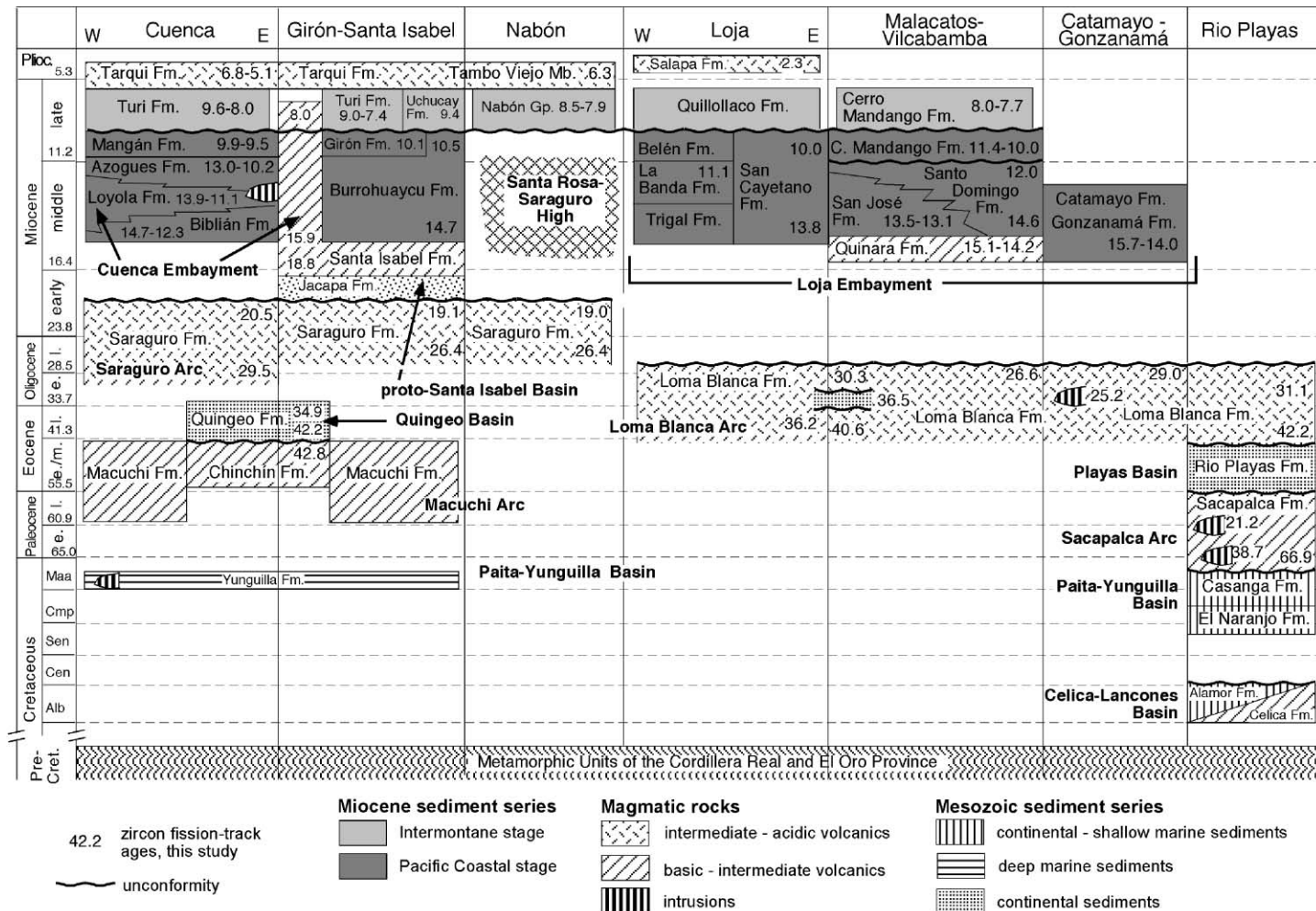


Fig. 13. Chronostratigraphic correlation chart of southern Ecuador. Fission-track key ages and ranges are indicated within the time scale of Berggren et al. (1995). For further discussion, see text.

vironments. In addition, rare benthic foraminifera species are present. Mollusc faunas reported from the Cuenca, Loja, Malacatos–Vilcabamba and Cata-mayo–Gonzanamá areas are dominated by freshwater taxa, but there are also species present that are known to tolerate changes in salinity that occur in marginal marine environments (Bristow and Parodiz, 1982; Nutall, 1990). Crab claws and shrimps in these sediments (Feldmann et al., 1993) provide further evidence for a marginal marine environment of deposition. In conclusion, the present combination of facies and faunal data implies that the lower sequence was deposited in a coastal plain and deltaic environment, which had significant freshwater input from various rivers in a tropical climate. The palaeogeographic position and clast types suggest that deposition of the lower sequence occurred between the Cordillera Real and the bordering Pacific ocean. Therefore, deposition of the lower sequence occurred during a *Pacific Coastal* stage and the chronostratigraphic ZFT data indicates that this stage lasted from about 15 to 9.5 Ma (Fig. 13).

The upper sequence generally comprises continental, i.e. alluvial fan and proximal fluvial facies elements, which are mainly capped by airborne volcanic formations. The main basal unconformity with the lower sequence is partly angular (Cuenca, Girón–Santa Isabel, Loja) although progressive unconformities are also observed in the continental formations (Vilcabamba, Nabón). Using our nomenclature, these sequences were deposited in the *Intermontane* stage basins, which formed in an $\approx$ ESE–WNW oriented compressional tectonic regime that resulted in uplift of the southern Ecuadorian realm (Steinmann, 1997; Hungerbühler, 1997; Steinmann et al., 1999). The compression and surface uplift restricted sedimentation to smaller areas, which approximately coincide with the perimeters of the outcrops of the Intermontane sequences. Continental sedimentation and uplift took place during the period from 9.5 to $\approx$ 6.0 Ma (Fig. 13), which is constrained by the following observations: (1) in the Cuenca area the Cojitambo intrusion, dated at 7.8 ± 0.8 Ma, cuts the steeply dipping sedimentary series of the Pacific Coastal sequence (Steinmann, 1997; Steinmann et al. 1999); (2) in the same area, the horizontally bedded volcanic Llacao Mb. (in its upper part dated at 5.1 ± 0.6 Ma) overlies the Mangán Fm. of the Pacific Coastal

sequence with a high angle unconformity (Steinmann, 1997); (3) the flat lying Uchucay Fm. (9.4 ± 1.6 Ma) overlies the deformed Burrohuaycu Fm. (14.7 ± 1.2 – 10.5 ± 1.2 Ma) with an angular unconformity reaching 40° in the Santa Isabel area; (4) the short-lived Intermontane sequence (8.5–7.9 Ma) in the Nabón area is characterized by numerous progressive unconformities, and the volcanic Tambo Viejo Mb. (6.3 ± 1.0 Ma) sealed the unconformities (Hungerbühler et al., 1995); (5) a progressive unconformity developed at ca. 8.0 ± 1.0 Ma (see above and Hungerbühler, 1997) in the Vilcabamba area in the Conglomerate Mb. (Cerro Mandango Fm.) of the Intermontane sequence; (6) the undated, but younger than ≈ 10 Ma Quillolaco Fm. seals the deformed Pacific Coastal sequence in the Loja area; and (7) modelling of apatite fission-track data acquired from the Pacific Coastal stage sequence of Cuenca suggests that exhumation commenced and the present day surface cooled through 60°C at ≈ 9.5 Ma (Steinmann, 1997; Steinmann et al., 1999). A calculated mean surface uplift of 0.3 mm/year brought the Pacific Coastal stage sequence to the today's altitude of approximately 2700 m (Steinmann et al., 1999) since ≈ 9.5 Ma in the Cuenca area. Mixed continental clastic and pyroclastic sedimentation occurred during the early Intermontane stage from ≈ 9.5 –8.0 Ma. Later, volcanic deposition prevailed (≈ 8.0 –5.0 Ma).

The present interpretation of late Miocene–Pliocene compression and uplift in the arc area corroborates with the thermochronological analyses of Spikings et al. (2000, 2001) from three traverses across the northern Cordillera Real of Ecuador. Apatite fission-track modelling in the metamorphic belt reveals that a substantial pulse of increased cooling has occurred since ≈ 10 Ma, implying compression-related uplift causing exhumation rates of up to 1.7 km/Ma. In contrast, during the preceding period between 15 and 10 Ma, which approximately correlates with the extension and Pacific Coastal stage (see discussion below) in the Inter-Andean region, no significant exhumation is observed in the Cordillera Real (Spikings et al., 2000). In conclusion, the late Miocene unconformity in the Inter-Andean region was created by the transition of extension to compression, which involved also the northern Andean chain of Ecuador.

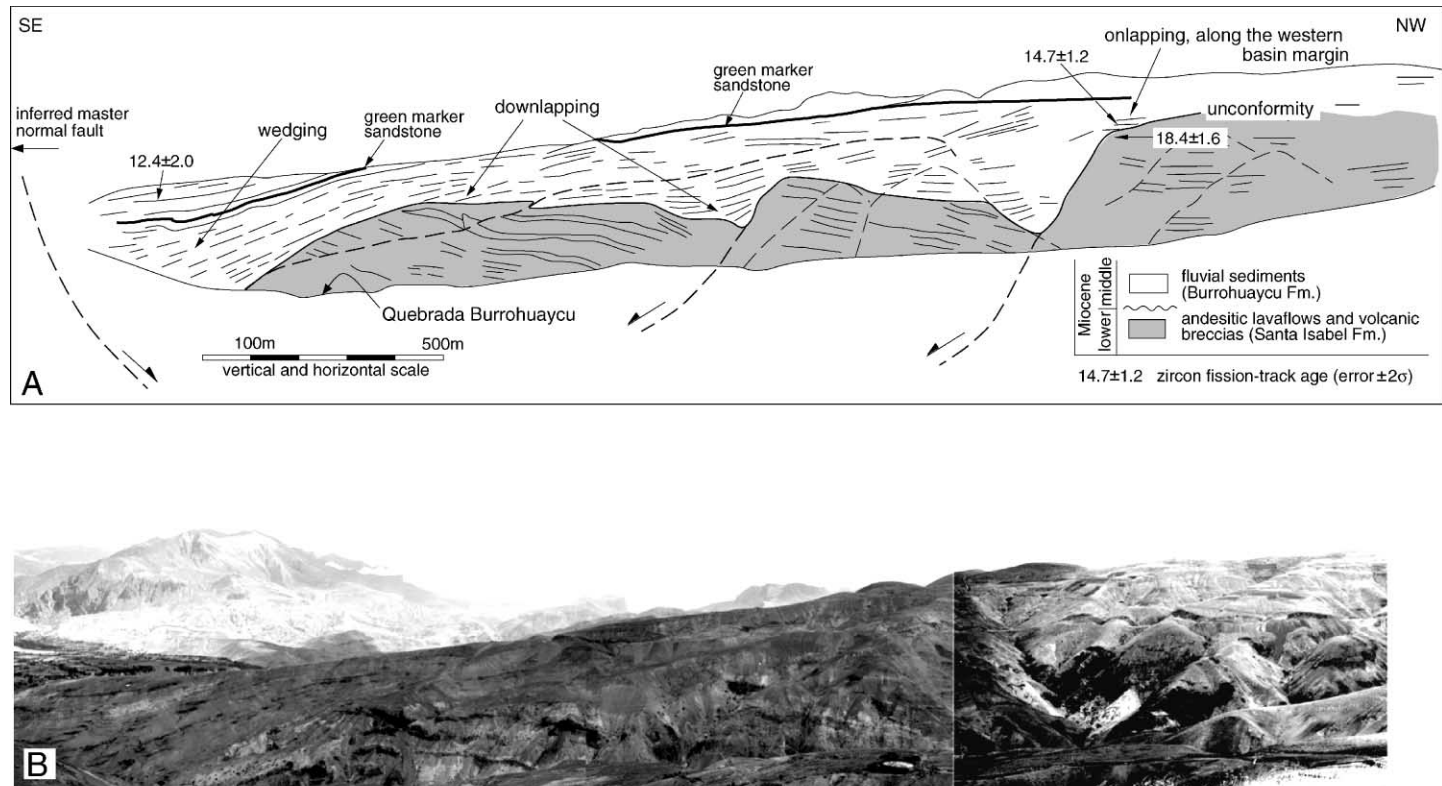


Fig. 14. Cross-section (A) and photograph (B) along the right slope of the Quebrada Burrohuaycu in the area south of Santa Isabel. The location of the section is indicated in Fig. 15. The Burrohuaycu Fm. was unconformably deposited on the block faulted volcanic Santa Isabel Fm. in a half-graben with an inferred SW–NE trending normal master fault to the SE. This synthetic normal fault was reversed during subsequent compression (see Fig. 15). From Hungerbühler (1997) and Helg (1997).

4. Tectono-sedimentary history: basin formation and inversion

The predominant young (< 9 Ma) deformation style in the Miocene basin series of southern Ecuador is compressional. Only few of the extensional structures were preserved in the middle Miocene (Pacific Coastal stage) sediments. However, in several places, the geometry of sedimentary formations, local bedding geometries and the presence of normal faults in the underlying volcanic formations indicate that extension predated the compression. In the following sections we shall explore these points in more detail.

4.1. Santa Isabel area

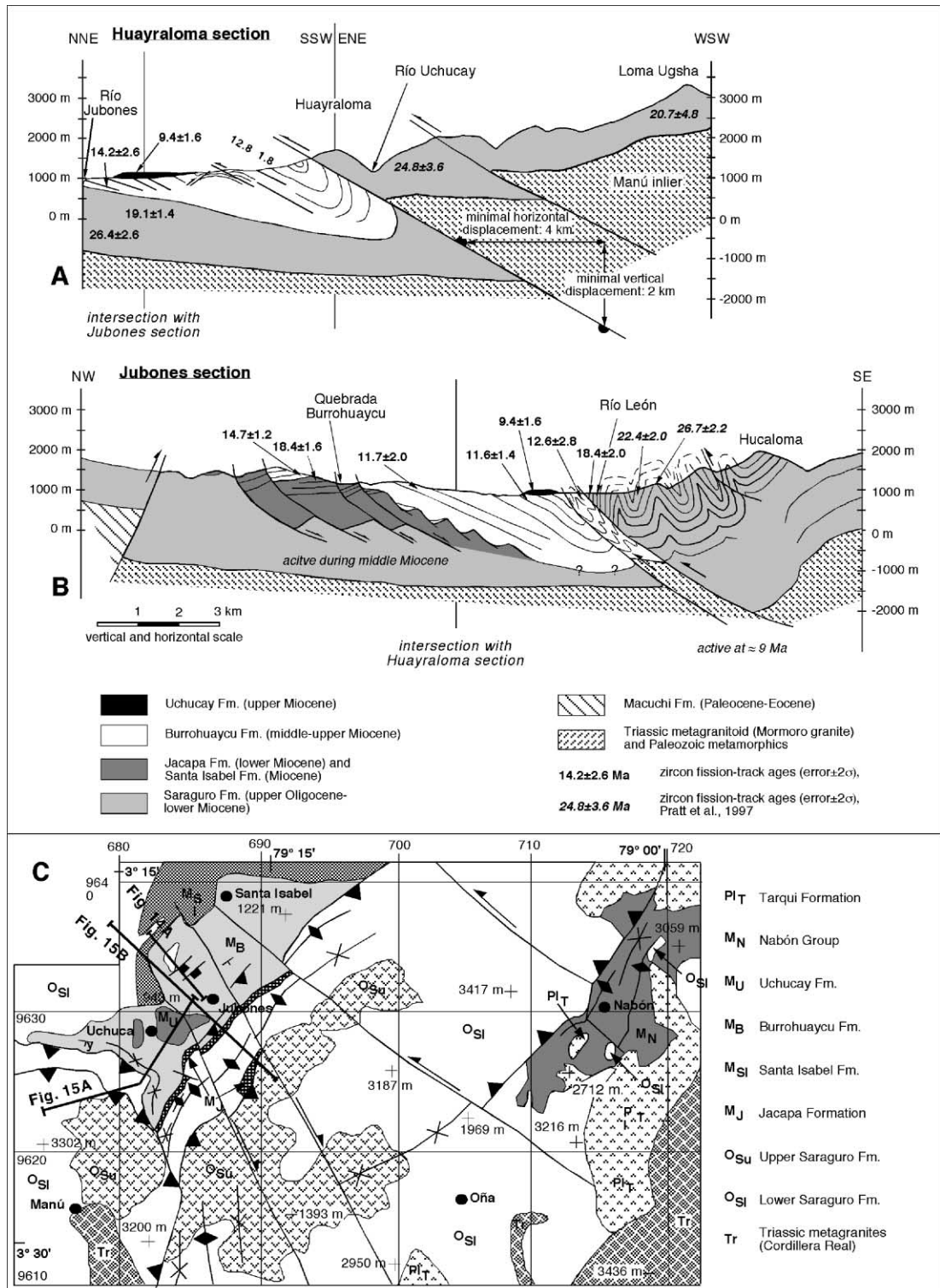
In the Santa Isabel area, and in particular in the Quebrada Burrohuayacu (south of Santa Isabel), NW–SE oriented extension during basin formation can be observed (Fig. 14). Hectametric, SE dipping normal fault blocks are exposed in the Santa Isabel Fm., which underlies the Burrohuayacu Fm. Their pre- and syn-sedimentary character is indicated by down and onlapping geometries of Burrohuayacu Fm. beds with the tilted volcanics. A prominent green marker sandstone bed in the Burrohuayacu Fm. forms a useful reference horizon (Helg, 1997) (Fig. 14). Thickening of the Burrohuayacu Fm. to the SE continues for several kilometers in the same direction (Hammer, 1998). This wedge-shaped geometry suggests the presence of a distant $\approx$ NW dipping normal master fault, and the smaller normal faults depicted in Fig. 14 form shallower antithetic structures with respect to the inferred crustal scale master fault (Hungerbühler, 1997; Helg, 1997; Hammer, 1998). However, the extensional master fault is no longer preserved, but was most likely rotated and inverted during subsequent compression that was parallel to the earlier NW–SE oriented extension.

This arises from regional mapping of the Huayraloma and Jubones sections (Hungerbühler, 1997; Pratt et al., 1997b). Fig. 15 displays the post-compressional scenario that is sealed unconformably by the upper Miocene Uchuay Fm. The basin flooring Saraguro Fm. overthrusts the asymmetric Burrohuayacu Fm. along a large scale thrust fault, creating a large footwall syncline in the proximal part and low amplitude folds in the distal part (Fig. 15A). Mild deformation observed in the latter part may explain why the pre- and syn-sedimentary normal faults were not inverted during general compression (Fig. 15B). The Saraguro Fm. is deformed in a 5-km-wide, N–S trending belt of chevron folds (Fig. 15B), indicating that deformation occurred during shallow burial. The chronostratigraphic fission-track ages prove that the compressive deformation occurred at ≈ 10 Ma as documented by the age of the unconformably overlying Uchuay Fm. (9.4 ± 1.6), which belongs to the Intermontane stage of basin development.

4.2. Loja area

A complex pattern of lithologic formations, geometries and detrital supply occurs in the Loja area. In particular, in the western and eastern regions the sedimentary sequences developed differently during the Pacific Coastal stage (see above), and they had different source areas. This observation requires an earlier, clear geographical separation of these sedimentation sites whereas during the later Intermontane stage, both the western and eastern sequences were tectonically assembled and unconformably overlain by the Intermontane Quillollaco Fm. (see Fig. 16). The eastward thickening of the Lower Sandstone Mb. (San Cayetano Fm.) in the eastern compartment implies that a wider half-graben was forming during initiation of extension during the Pacific Coastal stage (Fig. 16A). Continued exten-

Fig. 15. Two cross-sections in the Santa Isabel area. Note that the sections intersect each other and the locations are given in the geological map (C) (after Hungerbühler, 1997; Pratt et al., 1997a,b; Hammer, 1998). The cross-sections display large scale thrusting of the late Oligocene–early Miocene Saraguro Fm. over the middle Miocene Burrohuayacu Fm. during basin inversion, which occurred at ≈ 9 Ma. Deformation in the Burrohuayacu Fm. is sealed by the horizontally lying Uchuay Fm. In section (A) a minimum vertical and horizontal shortening along the thrust fault of about 2 and 4 km can be estimated. The chevron folding in the Saraguro Fm. in section (B) indicates an overall shortening of $\approx 30\%$ in the belt.



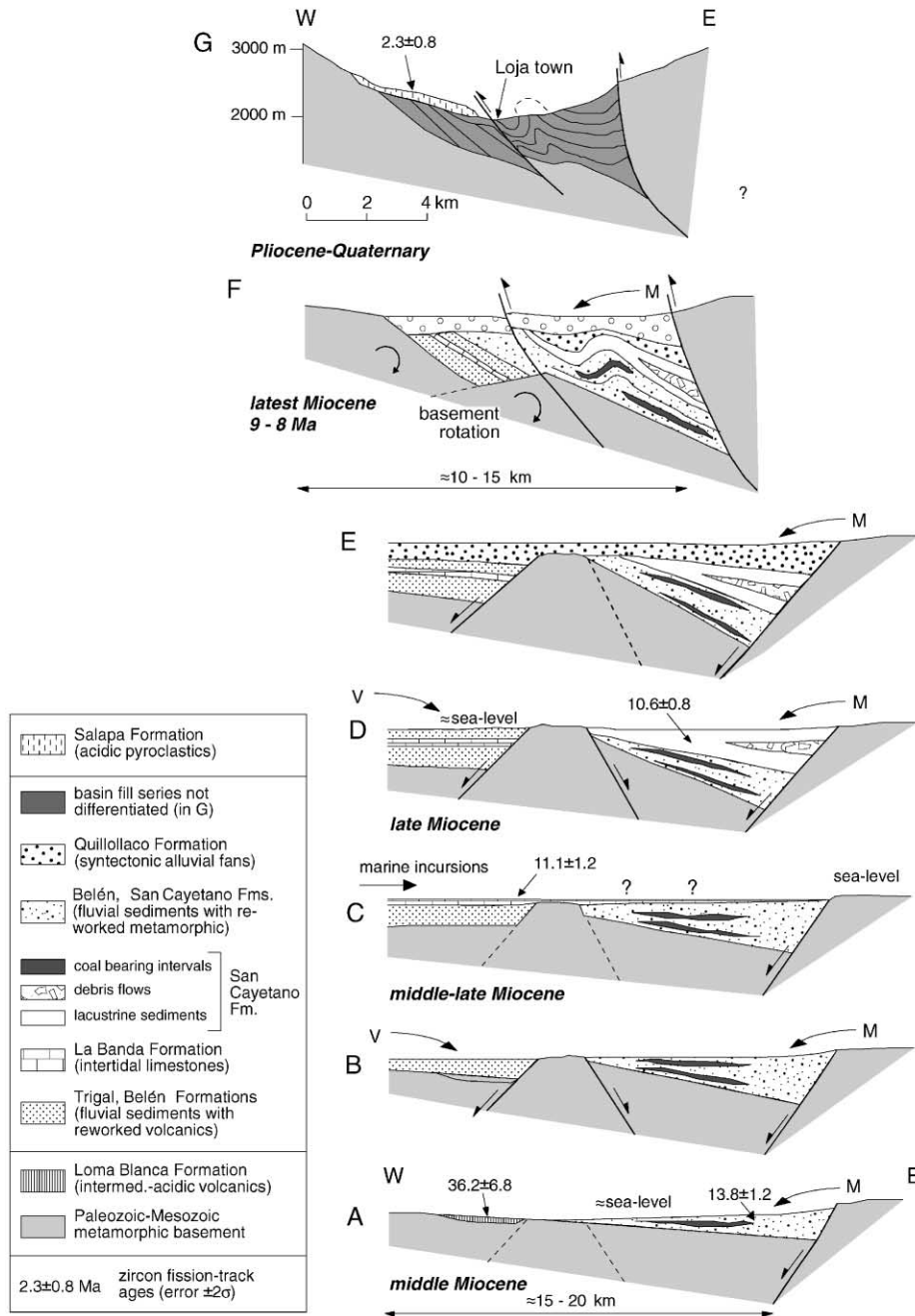


Fig. 16. Tectono-sedimentary evolution of the Loja area during the middle Miocene to the Pliocene. M = metamorphic clasts, V = volcanic clasts. Figures (A) to (E) depict the inferred extensional history during the Coastal Pacific stage, allowing for deposition of different sequences at about sea level in the eastern and western graben/half-graben, which were most likely separated by a horst block of unknown dimension. (F) shows the Intermontane stage situation during subsequent compression. Reverse faulting and basement rotation drove the tectonic assembling of the former half-graben/graben fill series, and the unconformable cover by the Intermontane sequence (Quillollaco Fm.). Compressional deformation and erosion prevailed until Pliocene and the youngest volcanic formation known in southern Ecuador (Salapa Fm.) sealed the erosional relief (G).

sion may have resulted in the formation of another half-graben to the west, which was perhaps separated from the eastern one by an emergent horst block. The western graben-filling Trigoal Fm. was supplied with volcanic clasts (Fig. 16B) and the conformably overlying La Banda Fm. marks a marine ingression from the west into the western graben (Fig. 16C). The ingression is not documented in the eastern graben and it may correlate with the unconformity observed between the Lower Sandstone and the Siltstone Mbs. of the San Cayetano Fm. (Fig. 8). Either the marine sediments were deposited and later eroded, or the unconformity represents the correlative base level in the eastern half-graben segment. Subsequently, the western and eastern basins were once again separated (Fig. 16D). Rapid subsidence in the eastern graben is indicated by massive debris flows and slumps in the otherwise fine-grained lacustrine succession (Siltstone Mb. of San Cayetano Fm.), which thickens to the east. In the western graben, delta plain and fluvial deposits (Belén Fm.) successively occupied the former lagoon (La Banda Fm.). In the east, the lake was filled by the Upper Sandstone Mb. of the San Cayetano Fm., which probably overstepped the separating ridge as suggested by the coeval Belén Fm., which contains metamorphic clasts in addition to volcanic material (Fig. 16E). The western graben was essentially tilted under the load which developed by the inversion of the former antithetic normal fault (Fig. 16E,F). The coarse alluvial fan and braided river sediments of the Quillollaco Fm. were deposited on the sequences of the Pacific Coastal stage with angular unconformities (Fig. 16F). The unconformity reflects the inversion of the eastern half-graben structure, which resulted in tight and partly isoclinal folding of the Pacific Coastal stage sediments with fold axes parallel to the strike of the newly formed reverse faults. Unfortunately, no chronostratigraphic control is available from the Quillollaco Fm. However, by comparison with the other sites in southern Ecuador, it is assumed that the coarse clastics prograded unconformably from the east over the deformed and partly eroded Pacific Coastal sequence at about the same time (≈ 10 – 8 Ma) (Fig. 16F). The E–W-directed compression continued and deformed the Intermontane sequence, giving rise to open synclines in this young sequence (Fig. 16G).

4.3. Cuenca area

The tectonic regime that controlled the deposition of the Pacific Coastal sequence in the Cuenca area can be derived from several circumstantial arguments. The presence of voluminous deltaic facies implies that one, or several large river systems entered from the east a body of standing water. The presence of metamorphic clasts implies that the incipient Cordillera Real was the dominant clastic source. Brackish and marine ostracods and other marginal marine faunal indicators suggest that a marine-based delta existed. The large thickness (1500–2500 m) of the sequence was deposited over a time span of ≈ 5 Ma corresponding to high accommodation rates of 300–500 m/Ma. During deposition of the Coastal Pacific stage, there is no evidence for the existence at this latitude of the Cordillera Occidental as a clastic source, or that it may have sheltered the area from the Pacific ocean. We suggest that the lower sequence of the Cuenca area was deposited in a dominantly extensional tectonic setting situated between the incipient Cordillera Real and the Pacific Ocean. Previously, Noblet et al. (1988) and Lavenue et al. (1995) assumed a transpressive strike-slip regime during formation and fill of the Cuenca Basin. This was based on the likely occurrence of progressive unconformities in the Azogues and Mangán Formations. According to our field mapping, this structure represents a footwall syncline due to post-sedimentary, E–W oriented thrusting (Steinmann et al., 1999). However, we agree that on a regional scale, the tectonic development of the Cuenca area was controlled by major right-lateral strike slip movement in the forearc area as suggested by Noblet et al. (1988), Noblet and Marocco (1989) and Marocco et al. (1995). In our opinion, efficient strain partitioning may have worked during the formation and inversion of the basins in general.

The tectono-sedimentary evolution of the Cuenca and adjacent regions (Girón, Nabón) is interpreted according to Steinmann (1997), and it is sketched in a series of palinspastic maps in Fig. 17 (see also Fig. 11). Following an erosional period during 20–15 Ma in the region, the deposition of the lower, Pacific Coastal sequence started within the meandering river system of the Biblián Fm. (Fig. 17A), which partly

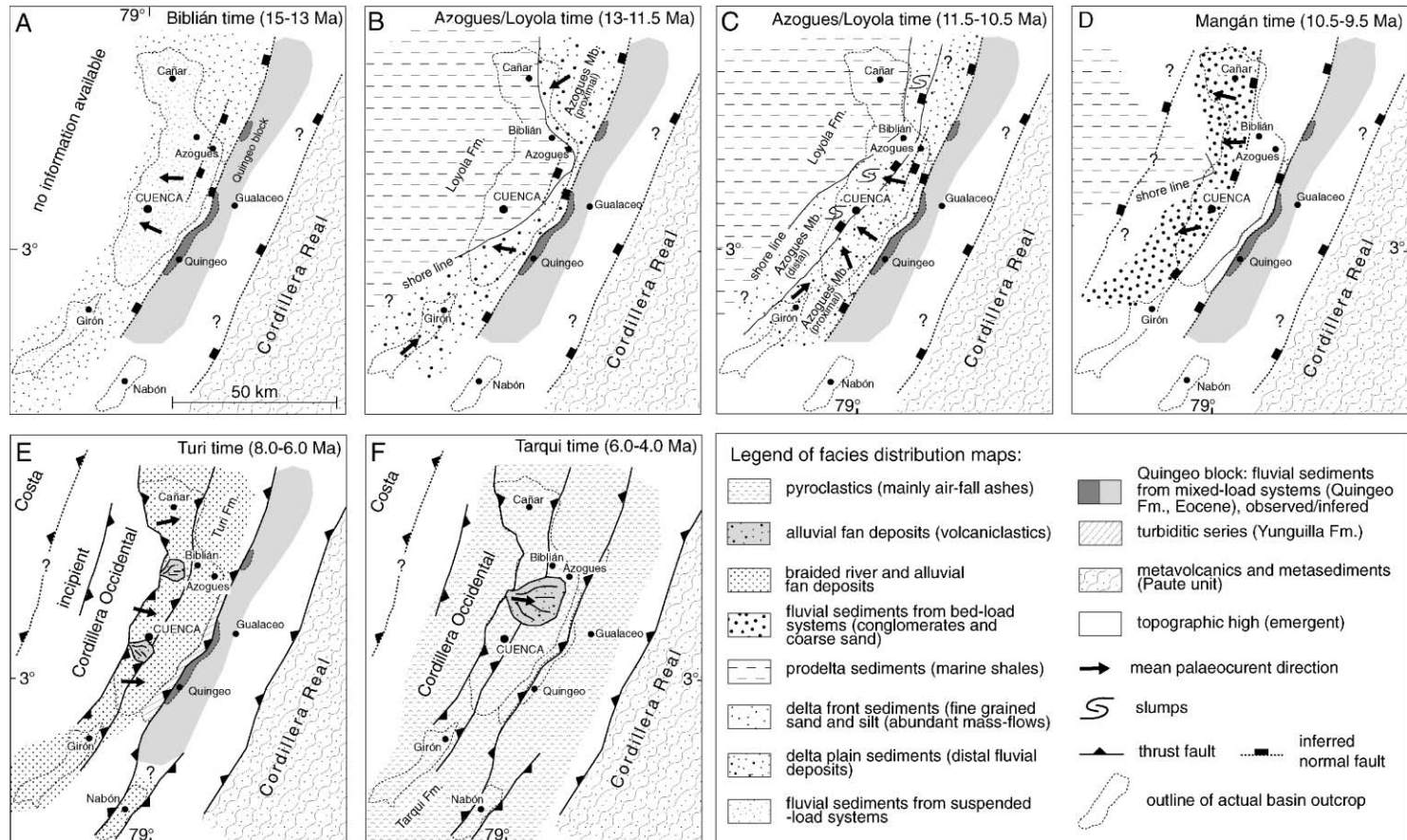


Fig. 17. Tectono-sedimentary evolution of the Cuenca area as derived from mapped facies distributions during middle-late Miocene time slices. Note that the Eocene Quingeo Fm. from ≈ 18 to 9 Ma experienced cooling (apatite fission-track modelling; Steinmann, 1997; Steinmann et al., 1999) while subsidence and sedimentation occurred to the west of the Quingeo block. This correlates with the Pacific Coastal stage in the Cuenca Embayment with prodelta, delta and generally westward prograding fluvial deposition (A–D). Compression and basin inversion during the late Miocene–early Pliocene (E–F) is documented by unconformable deposition of the continental Intermontane formations on folded Pacific Coastal stage formations, the first supply of volcanic material from the rising Cordillera Occidental, and the surface uplift and cooling (since ≈ 9 Ma, apatite fission-track modelling; Steinmann, 1997; Steinmann et al., 1999) of the Pacific Coastal sequence. Coeval overthrusting of the Quingeo block (deformed in a spectacular footwall-syncline) by the Yunguilla Fm. and overlying Coastal Pacific stage formations can be interpreted from apatite fission-track modelling (Steinmann, 1997; Steinmann et al., 1999), which suggests that the Quingeo Fm. remained in an about isothermal position from ≈ 9.5 to 4 Ma, prior to the final surface uplift to the present altitude. Modified from Steinmann (1997).

onlapped the folded Yunguilla Fm. to the east. Apatite fission-track modelling suggests that the Quingeo Fm., presumably together with the underlying Chinchín and Yunguilla Fms. at the eastern margin of the basin, cooled during 18–9 Ma through the partial annealing zone (≈ 110 –60 °C; Steinmann, 1997; Steinmann et al. 1999). The depotcenters of the Biblián Fm. and younger formations of the Pacific Coastal sequence must have subsided with respect to the eastern margin. Thus, a main eastern basin bounding normal fault is implied (Fig. 17A–D), which separated the cooling footwall block to the east from the subsiding basin on the hanging wall to the west. As the detrital material in the Biblián Fm. was supplied from the Cordillera Real, a normal fault stepped morphology further towards the Cordillera Real can be inferred. Continued general subsidence caused a marine ingression from the west during Loyola time (Fig. 17B). The Loyola Fm., along with the interfingering Azogues Mb., built the initial delta front and prodelta complex, which prograded towards the west. In the larger southern Cuenca area the main delta, which was partly fed by a river system draining the Girón area, developed during Azogues time (Fig. 17C). This was followed by a general westward shift of the depot center due to increased subsidence in the west during the time of deposition of the Mangán Fm. This may have been controlled by the activation of a new normal fault striking from Girón in the south to Biblián in the north (Fig. 17D). As a consequence, the former Azogues delta was overstepped by the delta plain and fluvial deposits of the Mangán Fm. Exposures of off-shore prodelta sediments of the Mangán Fm. are sparse (Fig. 17D). Modelling of apatite fission-track data (Steinmann, 1997; Steinmann et al. 1999) suggests that sediment burial heating occurred during deposition of the thick Pacific Coastal sequence. A major tectonic inversion during 9.5 to 8 Ma is modelled, coeval with the start of the Intermontane sequence deposition. This is documented by: (1) development of the angular unconformity between the Pacific Coastal and Intermontane sequences (Turi and Tarqui Fms.) in the Cuenca and Girón areas; (2) surface uplift and thrusting of the Cordillera Occidental started supplying detritus to the Cuenca area (Santa Rosa Mb. of the Turi Fm., Fig. 17E,F) at this time, and (3) the establishment and filling of the

compressive, short-lived Nabón Basin (Fig. 17E). Finally, the Tarqui Fm. (mainly Tambo Viejo Mb. airborne volcanics) capped the morphology of the Inter-Andean region and eastern parts of the Cordillera Occidental (Fig. 17F).

5. Regional tectonic implications

The present paper focuses on the Neogene history of Ecuador, following the break-up of the Farallon Plate into the Nazca and Cocos plates at ≈ 25 Ma (Pilger, 1984; Pardo-Casas and Molnar, 1987). However, we also present age and stratigraphic data from Paleogene volcanic and sedimentary formations that permit preliminary interpretations of the early Tertiary history. The interpretations mainly rely on varying convergence rates and vectors between the South American margin (SOAM) and the subducting oceanic plates. From the Eocene to the early Miocene, a continental arc with abundant volcanic activity prevailed in the Inter-Andean region as recorded by the widespread Chinchín Fm. and Saraguro Group (Loma Blanca and Saraguro Fms.). Only a few dominantly sedimentary series are preserved from this time (Río Playas and Quingeo Fms.). During the late Paleocene and Eocene periods, repeated deformation occurred in the Ecuadorian forearc (Jaillard et al., 1995). The middle to late Eocene was also a period of rapid, oblique (NE–SW oriented) convergence at the SOAM (Pardo-Casas and Molnar, 1987; Daly, 1989) and the formation of the Quingeo Fm. may have been the response to this event. Clastic supply to the Quingeo Basin from the Cordillera Real (Steinmann, 1997) during a period of increased exhumation in the Cordillera Real (≈ 43 –30 Ma; Spikings et al., 2000, 2001) indicates that coeval surface uplift was occurring in the eastern part of the Ecuadorian Andes. The Oligocene to early Miocene period was characterized by generally low and oblique convergence at the Ecuadorian subduction system (Pilger, 1984; Pardo-Casas and Molnar, 1987) with trench-normal extension in the forearc (Daly, 1989). This extension most likely stepped over into the Inter-Andean arc and may have driven the voluminous ignimbritic eruptions of the Loma Blanca and Saraguro Fms.

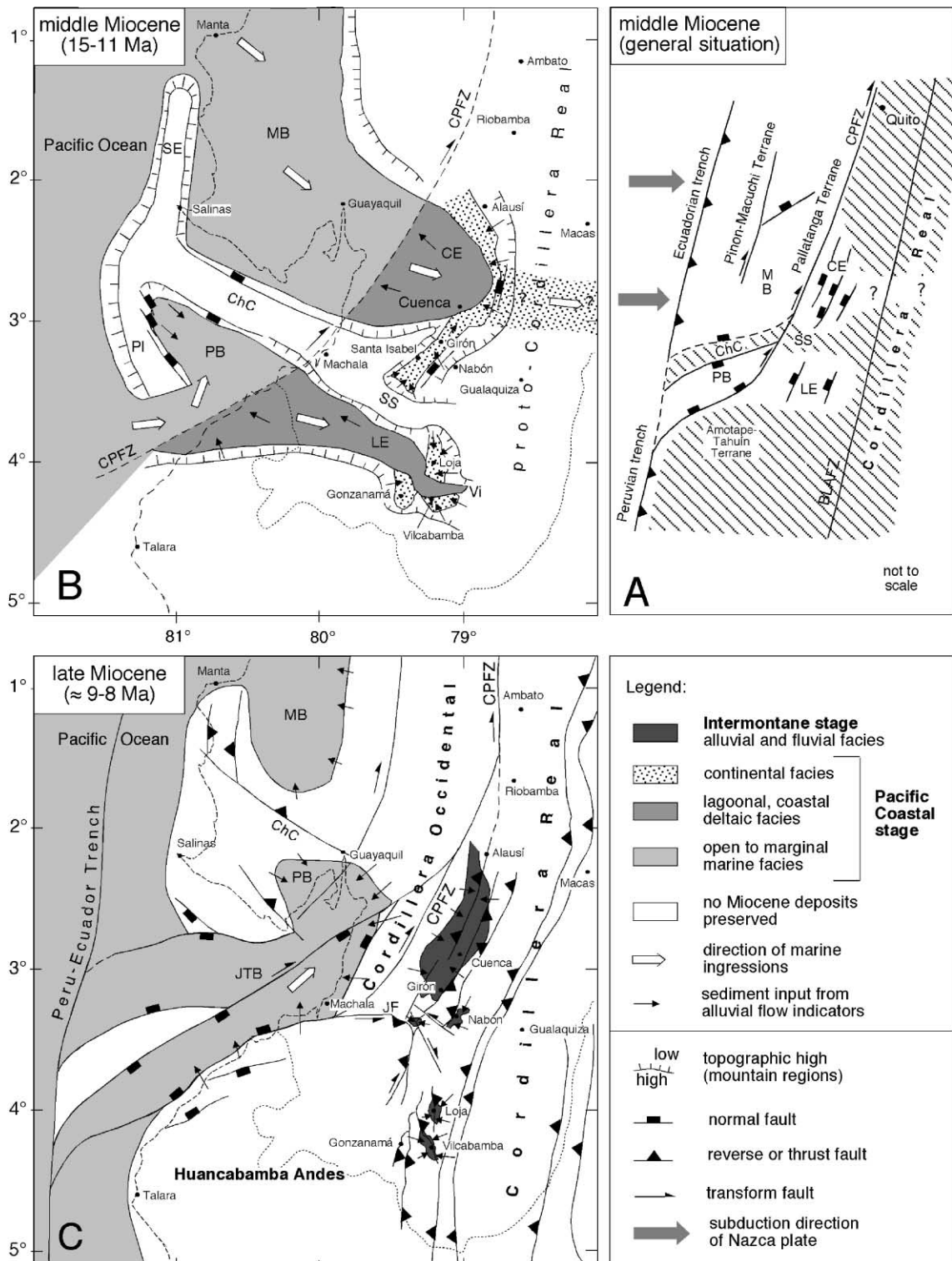
The Neogene history of the Inter-Andean region can be integrated with the development of the southern Ecuadorian forearc area and in particular with the tectonic displacement along the Calacali–Pallatanga Fault zone (CPFZ, Aspdén et al. 1995; Hughes and Pilatasig, 1999). The CPFZ is the main tectonic divide between the consolidated older Andean arc (Cordillera Real and Interandean Valley), which formed during early Cretaceous (≈ 140 – 120 Ma, Litherland et al., 1994), and the oceanic and volcanic arc terranes (Pallatanga and Piñon–Macuchi Terranes, Hughes and Pilatasig 1999; Spikings et al., 2001) in the Cordillera Occidental and Costa areas, which were accreted during latest Cretaceous and Eocene. The term CPFZ replaces the poorly defined Dolores–Guayaquil–Megashear (e.g. Lebrat et al., 1987) and similarly represents a dextral Inter-Andean shear zone that joins the Peruvian trench to the south of the Gulf of Guayaquil (Fig. 18A). The CPFZ has been reactivated since the Miocene and a right lateral displacement of ≈ 100 – 130 km between the Pallatanga and Piñon–Macuchi Terranes with respect to the South American continent has been estimated (Shepherd and Moberly, 1981; Hungerbühler, 1997). This displacement supposedly has affected the entire forearc area by driving subsidence and emergence of several pull-apart basins in the northnortheastward-displacing Piñon–Macuchi block, as well as to the south between the Amotape Tahuin block. These are the middle to late Miocene Manabí, Progreso and Jambelí–Tumbez Basins (Marks, 1951; Baldock, 1982; Daly, 1989; Deniaud et al., 1999; Deniaud, 2000; Fig. 18). The Jambelí–Tumbez basin opened last during the transition from the middle to the late

Miocene (Benítez, 1986, 1995). During their early development, these basins were separated by sub-aerial ridges (e.g. Chongón–Colonche High) and hosted very thick sequences in the range of 4 km (Manabí, Progreso) to 12 km (Jambelí–Tumbez).

A restoration of the middle Miocene situation by moving the coastal block back for 100–130 km (keeping the orientation of the controlling faults fixed) juxtaposes the Manabí Basin with the Cuenca/Girón–Santa Isabel area and the Progreso Basin with the Loja, Catamayo–Gonzanamá and Malacatos–Vilcabamba areas (Fig. 18A). The Chongón–Colonche High (Benítez, 1995), located between the Manabí and Progreso Basin, was most likely situated to the west of the Santa Rosa–Saraguro High, which in turn separated the corresponding depositional areas in the Inter-Andean domain (Fig. 18A). To constrain this palinspastic reconstruction it is necessary to determine if the facies development in these forearc basins fits with the proposed middle to late Miocene Inter-Andean sedimentary history. In the following, the chronostratigraphic correlation scheme of Berggren et al. (1995) is applied.

Within the Progreso Basin, above the middle Miocene (≈ 15 – 14 Ma) erosional unconformity, subsidence and transgression occurred from ≈ 14 – 11 Ma. The middle to upper Miocene Progreso Fm. transgressed eastward, depositing shallow marine to brackish sediments (Whittaker, 1988) over older shelf and coastal formations (lower Miocene Dos Boscas Fm.; Kennerley, 1980). The upper Progreso Fm. contains only a very sparse foraminifera fauna (Thalman, 1946). The basin area became emergent in the late Miocene (≈ 11 – 5 Ma; Benítez, 1995). In the

Fig. 18. Proposed palinspastic reconstruction of southern Ecuador during middle to late Miocene. In (B) and (C), the present coast lines are shown. Abbreviations: BLAFZ (Baños–Las Aradas Fault Zone), ChC (Chongón–Colonche High), CE (Cuenca Embayment), CPFZ (Calacali–Pallatanga Fault Zone), JF (Jubones Fault), JTB (Jambelí–Tumbez Basin), LE (Loja Embayment), MB (Manabí Basin), PI (Playas High), PB (Progreso Basin), SE (Santa Elena High), SS (Santa Rosa–Saraguro High). (A) Regional schematic reconstruction of the middle Miocene tectonic situation. The right-lateral displacement of the Piñon–Macuchi Terrane along the Calacali–Pallatanga Fault Zone drove subsidence in the forearc Manabí and Progreso pull-apart basins, which were separated by the Chongón–Colonche High, most likely juxtaposed to the Santa Rosa–Saraguro High in the arc region. This caused the removal of crustal support and extensional collapse in the Inter-Andean arc region. (B) Consequently, marine incursions occurred in the Cuenca and Loja Embayments where deltaic and fluvial systems entered and filled the shallow sea from the east. From the Cuenca Embayment, a connection across the early Cordillera Real with the Pebas depositional system (Amazon Basin in Peru and Brasil) may have existed. (C) In late Miocene general E–W compression started, but continued northnortheastward movement of the forearc units caused coeval subsidence of the Jambelí–Tumbez Basin. In the Inter-Andean (arc) region, surface uplift and continental sedimentation occurred in geographically limited compressive basins. Modified from Hungerbühler (1997).



neighboring and newly developing Jambelí–Tumbez Basin, the very thick late Miocene Progreso Fm. (≈ 11 –5 Ma) was deposited in a deltaic–estuarine environment (Benítez, 1986). In the Manabí Basin to the north, early and middle Miocene (≈ 23 –14 Ma) open marine sedimentation prevailed (Deniaud, 2000) and a shallowing towards the east is observed (Whittaker, 1988). Above a late middle Miocene hiatus (≈ 14 –13 Ma), the Manabí Basin sedimentation shows from the late Miocene to the earliest Pliocene (≈ 11 –5 Ma) a shallowing-upward sequence from upper bathyal to lagoonal facies (Portoviejo Fm., Daule Group, Jama and Canoa Fms.; Whittaker, 1988; Deniaud et al., 1999).

In the limits of error, the geochronologic correlation of the forearc basins with the Inter-Andean basins suggests that the sedimentary facies in the forearc region may have represented deeper and marginal marine correlatives of the Pacific Coastal stage facies in the Inter-Andean domain. This is consistent with the general observation that in the middle Miocene forearc basins, the depot centers migrated eastward towards the continent (Benítez, 1995) and may have been linked with the eastward-directed marine ingressions into the Inter-Andean Cuenca and Loja Embayments (15–11 Ma; Fig. 18B). Subsequently, the middle/late Miocene transition, coinciding with the change from the Pacific Coastal to the Intermontane stage (≈ 10 –9 Ma) in the Inter-Andean region, is characterized by: (1) a hiatus and renewed shallowing-upward sequence in the Manabí Basin; (2) the cessation of sedimentation and emergence in the Progreso Basin domain in the late Miocene; and (3) establishment of a deltaic estuarine system in the newly subsiding Jambelí–Tumbez Basin to the south.

There are also tectonic arguments, which suggest that the sedimentary development in the forearc was connected with the Inter-Andean region. The middle Miocene NNE–SSW-oriented extension in the forearc domain, driven by the right-lateral displacement of the coastal block along the Calacali–Pallatanga Fault zone (Aspden et al., 1995), has thinned the underlying crust in the forearc area. Consequently, the Inter-Andean region lost lateral crustal support in the west, and extensional collapse utilising NNE–SSW trending normal faults may have taken place (Fig. 18A), allowing marine ingressions from the

Pacific side and the deposition of the Pacific Coastal sequences. This occurred in two individual domains north and south of the separating Santa Rosa–Saraguro High, which we refer to as the Cuenca and Loja Embayments (Hungerbühler, 1997; Fig. 18B). Subsequent late Miocene compression in the forearc (Daly, 1989) most likely stepped over into the Inter-Andean region. The former marine embayment basins were inverted and several continental Intermontane stage basins established. Coeval displacement of the forearc along the Calacali–Pallatanga Fault zone drove the subsidence in the Jambelí–Tumbez Basin.

The full integration of the Neogene sedimentary evolutionary sequence into a larger plate tectonic scheme for the Northern Andes and Pacific is still difficult to compile. It appears that the tectonic and magmatic evolution of the Andean margin is controlled by the complex interplay of changing rates of subduction, obliquity and angle of subduction, subduction roll-back and the changing spreading rates in the Equatorial Atlantic (see e.g. Aspden et al., 1987; Daly, 1989; Sébrier and Soler, 1991). The timing of these factors is still poorly understood although since the breakup of the Farallon plate at ≈ 25 Ma, the obliquity of subduction of the Nazca plate under the South American plates has been relatively constant (approximately towards the E) and subduction rates have been high (≈ 100 mm/year; ; Pardo-Casas and Molnar, 1987). Therefore, middle Miocene extension in the Inter-Andean area appears to have occurred during high rates of plate convergence (120 ± 35 mm/year; Pardo-Casas and Molnar, 1987). The proposed effect of removing crustal support in the Inter-Andean region via the NNE-directed displacement of the coastal block may be an appropriate explanation for middle Miocene extension in the Inter-Andean region. However, there is still no adequate explanation for the cause of compression in the Inter-Andean region, which started at ≈ 10 –9 Ma. It may be speculated that the subduction of the Carnegie Ridge with the Ecuadorian margin has strongly controlled the Neogene tectonic development of the forearc and arc areas. The underthrusting of thickened buoyant oceanic crust results in plate coupling and compression and uplift in the overriding continental margin.

Traditionally, the collision between the Carnegie Ridge and the SOAM was thought to have occurred

at 1–3 Ma (e.g. Lonsdale, 1987). Pilger (1984) and Daly (1989) tentatively suggested that the subduction of the aseismic Carnegie Ridge may have driven compression in Ecuador since the late Miocene (≈ 10 Ma). The recent recognition that the Carnegie Ridge extends 400–500 km inland of the Ecuadorian trench under the Ecuadorian Andes implies that it collided at ≈ 8 Ma or earlier with the South American continent (Gutscher et al., 1999). Plate convergence calculations for the last ≈ 22 Ma (the minimum reference starting time for eastward movement of the Carnegie Ridge) were carried out by Spikings et al. (2001). By utilising the minimum and maximum convergence rates, the authors propose that the Carnegie Ridge collided with the Ecuadorian trench and forearc at some time between 15 and 9 Ma (Spikings et al., 2001). The maximum estimate coincides with the start of the dextral NNE-directed displacement of the coastal block with respect to the SOAM and the formation of the Manabí and Progreso pull-apart basins over the coastal region. Crustal collapse in the arc region may have first allowed marine ingressions into the Loja and Cuenca Embayments (Pacific Coastal stage). By ≈ 9 Ma, continued subduction and coupling of the buoyant plateau may have increased compression as manifested by (1) emergence and shallowing upward in the forearc basins (Progreso, Manabí) and (2) compression, surface uplift and continental deposition (Intermontane stage) in the newly developed Inter-Andean region. Coupling also should have enhanced the NNE displacement of the coastal block with rapid subsidence in the Jambelí–Tumbez basin since ≈ 11 Ma. Alternatively, a later collision of the Carnegie Ridge (≈ 10 –9 Ma) may have driven general compression in the forearc and arc. However, in our opinion the start of collision at ≈ 15 Ma offers a more appropriate explanation for the succession of events in time and space.

6. Conclusions

We have compiled chronostratigraphic ages and sedimentologic and faunal observations from the Tertiary sedimentary and volcanic formations, which have permitted a detailed model for the Neogene sedimentary and tectonic history of the southern

Ecuadorian Andean domain. The proposed two main stages of middle Miocene and late Miocene development are illustrated in Fig. 18. The model integrates regional tectonic aspects but differs in part from earlier reconstructions. These differences have been derived by providing a detailed chronostratigraphy of the sedimentary sequences, and by presenting arguments for middle Miocene marine ingressions from the Pacific ocean, which reached at least the Inter-Andean region.

During the middle Miocene Pacific Coastal stage (from ≈ 15 to 10 Ma), extensional subsidence in the Inter-Andean region caused an ingression of shallow seas into the Cuenca and Loja Embayments from the Manabí and Progreso Basins (Fig. 18A). The Santa Rosa–Saraguro High operated as an embayment divide, which most likely had its western continuation in the Chongón–Colonche High in the forearc domain. In both realms, the delta systems were fed by rivers that mainly sourced in the Cordillera Real, and adjacent, ancient and coeval volcanic series. The Girón–Santa Isabel area remained in a continental position and drained towards the Cuenca Embayment. Local extension may have stepped back inland, across the Cordillera Real, opening a connection with the Amazonian region represented by the Pebas sequence (Fig. 18B) (Hungerbühler, 1997; F. Wesselingh, personal communication, 2000). The deltas rapidly filled the embayments and in later stages, the fluvially dominated deltas held back several larger freshwater lakes. Decreasing subsidence and/or strong detrital supply from the embayments may have caused the general shallowing-upward sequence development in the connected yet distant Manabí and Progreso Basins.

Compression and tectonic inversion started in the forearc area and in the Inter-Andean area from 10 to 9 Ma. In the Inter-Andean region, several smaller Intermontane basins developed from ≈ 9.5 to 5.0 Ma (Cuenca, Girón–Santa Isabel, Loja, Vilcabamba, Fig. 18C) in which the older Pacific Coastal sequences were preserved below the Intermontane sequences. In adjacent uplifted areas, these older sequences were most likely eroded and partly tectonically buried and preserved below thrust sheets (e.g. Santa Isabel, Catamayo–Gonzanamá). The Nabón Basin formed rapidly during the period of strongest compression in the Inter-Andean region (≈ 9 –8 Ma).

Appendix A

Zircon fission-track dating results of volcanic and sedimentary formations in southern Ecuador (compiled from Hungerbühler et al., 1995; Steinmann, 1997; Hungerbühler, 1997).

REGION, Formation, Sample	Grid reference	No. of grains analysed	$\rho_d \times 10^4 \text{cm}^{-2}$	$\rho_s \times 10^4 \text{cm}^{-2}$	$\rho_l \times 10^4 \text{cm}^{-2}$	U (ppm)	Var. (%)	$P(\chi^2)$ (%)	Age $\pm 2\sigma$ (Ma)	Depositional age if different from bulk age	
CUENCA AREA											
Tarqui Fm.											
MS 218 (Llacao Mb.)	732155/9672200	18	47.72 (2913)	11.78 (289)	56.16 (1378)	47	0	91	23.2 $\pm$ 3.2	5.1 $\pm$ 0.6	
DH 213	729094/9687475	20	40.42 (5675)	62.89 (439)	228.2 (1593)	226	0	100	5.5 $\pm$ 0.6		
MS 216	731940/9672631	14	50.09 (5675)	148.0 (326)	639.7 (1409)	511	0	98	5.8 $\pm$ 0.8		
MS 234	728773/9664281	16	40.45 (5675)	86.80 (323)	300.7 (1119)	297	2	51	6.0 $\pm$ 1.0		
MS 414	720818/9672195	8	37.72 (3441)	80.16 (208)	253.2 (657)	262	3	44	32.4 $\pm$ 10.2	6.1 $\pm$ 1.0	
MS 235	729453/9664218	19	48.41 (5675)	91.86 (387)	353.2 (1488)	292	5	56	6.3 $\pm$ 0.8		
MS 233	728865/9663789	20	43.95 (2913)	62.52 (280)	213.0 (954)	194	0	100	13.4 $\pm$ 5.2		
DH 220	744998/9700442	20	45.42 (2913)	60.88 (454)	201.8 (1505)	178	1	75	6.7 $\pm$ 0.8		
MS 432	723285/9725220	19	34.87 (3441)	76.78 (487)	205.3 (1302)	230	0	92	6.8 $\pm$ 0.8	6.6 $\pm$ 0.8	
Cojitambo Intrusion											
MS 320	737503/9691772	14	34.76 (3441)	109.7 (569)	372.0 (1929)	417	0	98	5.4 $\pm$ 0.6		
MS 207	735010/9695790	12	49.61 (2913)	115.3 (514)	360.6 (1608)	291	0	97	7.8 $\pm$ 0.8		
Gualashi Intrusion											
MS 491	722784/9651739	11	41.63 (2792)	99.2 (294)	282.8 (838)	265	0	89	7.6 $\pm$ 1.0	7.6 $\pm$ 1.0	
Turi Fm. (Turi Mb.)											
MS 433	732196/9721603	20	39.74 (4173)	62.21 (452)	162.3 (1179)	163	0	95	8.0 $\pm$ 1.0		
DH 262	721221/9672528	19	35.21 (3441)	40.45 (283)	93.05 (651)	106	0	100	8.0 $\pm$ 1.2		
DH 219	743417/9697864	20	43.65 (6006)	823.1 (617)	20.88 (1565)	191	1	65	8.6 $\pm$ 0.8		
MS 391	717309/9668285	17	39.33 (4137)	348.4 (182)	76.57 (400)	76	0	100	9.4 $\pm$ 1.8	9.4 $\pm$ 1.8	
MS 283	722317/9671161	5	36.01 (3441)	203.9 (185)	402.3 (365)	447	0	94	9.6 $\pm$ 1.8		
Mangán Fm.											
MS 377	733213/9697750	16	35.78 (3441)	202.7 (699)	396.2 (1366)	443	1	69	9.5 $\pm$ 1.0	9.5 $\pm$ 1.6	
MS 342	732386/9689118	7	38.06 (3441)	15.49 (209)	32.31 (436)	331	0	99	29.4 $\pm$ 8.8		
MS 344	733053/9693300	7	37.38 (3441)	264.1 (270)	524.2 (536)	547	0	77	9.9 $\pm$ 1.6		
MS 405	732117/9690104	18	39.50 (4173)	73.18 (445)	152.4 (927)	151	0	100	9.9 $\pm$ 1.2		
Azogues Fm. (Cochas Mb.)											
MS 231	729696/9678349	20	44.20 (5675)	181.4 (1210)	386.5 (2578)	350	10	7	10.2 $\pm$ 1.2	10.2 $\pm$ 1.2	
MS 308	734372/9683168	16	36.81 (3441)	191.0 (555)	361.7 (1051)	393	4	77	10.2 $\pm$ 1.2		
MS 280	722391/9676437	14	38.63 (3441)	228.5 (528)	438.9 (1014)	443	1	46	10.6 $\pm$ 1.2		
MS 232	732352/9678437	17	43.36 (5675)	222.4 (806)	436.2 (1581)	402	0	100	11.0 $\pm$ 1.0		

Azogues Fm. (Guapán Mb.)

MS 319	739163/9696163	20	39.25 (4173)	90.67 (451)	162.6 (809)	162	0	100	11.5 ± 1.4
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Azogues Fm. (Azogues Mb.)

DH 209	733790/9687405	20	50.51 (5675)	205.6 (1581)	436.9 (3359)	346	0	89	11.9 ± 1.2
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DH 205	729883/9679424	20	51.77 (5675)	205.9 (1488)	425.4 (3074)	329	5	30	12.4 ± 0.8
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MS 211	728361/9675003	10	40.16 (5603)	192.2 (569)	305.1 (903)	279	8	29	12.6 ± 1.6
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DH 206	730935/9680283	23	47.43 (5301)	239.6 (1159)	424.1 (2051)	358	7	50	13.0 ± 1.0
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Loyola Fm.

MS 290	729893/9679345	20	34.98 (3441)	208.2 (925)	344.5 (1531)	394	0	86	11.1 ± 1.0
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MS 437	737662/9691463	20	36.69 (3441)	250.0 (1605)	429.5 (2757)	468	2	58	11.2 ± 0.8
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MS 274	723560/9676488	18	35.10 (3441)	225.3 (1022)	343.5 (1558)	391	0	91	12.1 ± 1.0
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MS 208	725275/9676852	21	41.68 (5675)	207.5 (1659)	356.6 (2851)	342	3	37	12.1 ± 1.2
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DH 208	734038/9688019	26	47.15 (5675)	328.8 (2806)	558.0 (4762)	473	4	25	13.9 ± 1.4
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Biblián Fm.

MS 261	725779/9675714	20	35.33 (3441)	118.9 (493)	179.4 (7443)	198	7	73	12.3 ± 1.6
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MS 209	725954/9676206	8	49.67 (5675)	269.0 (242)	446.9 (402)	360	0	72	14.7 ± 2.4
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Saraguro Fm.

MS 367	735092/9706274	15	38.17 (3441)	502.4 (981)	489.6 (956)	500	2	61	20.5 ± 2.0
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MS 365	735732/9702400	20	35.67 (3441)	678.8 (1717)	599.0 (1515)	672	0	98	21.2 ± 1.6
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MS 478	695744/9693134	19	41.27 (2792)	341.0 (1299)	321.0 (1223)	303	0	85	22.8 ± 2.0
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MS 364	735533/9699850	25	35.89 (3441)	323.0 (1843)	258.0 (1477)	288	0	100	23.2 ± 1.8
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MS 268	726472/9673198	18	36.92 (3441)	600.1 (1638)	449.8 (1226)	475	0	98	25.8 ± 2.2
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DH 240	730250/9675252	20	46.31 (5675)	605.4 (1992)	534.3 (1758)	462	0	99	26.0 ± 1.8
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MS 470	714257/9685269	17	40.68 (2792)	209.0 (1128)	167.1 (902)	160	0	91	26.4 ± 2.6
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MS 501	698991/9674203	20	41.75 (2792)	308.9 (1065)	234.0 (807)	224	0	99	26.6 ± 2.8
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MS 214	730307/9675184	6	43.74 (2913)	340.5 (356)	267.8 (280)	244	0	97	27.1 ± 4.4
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DH 241	730062/9675245	9	42.94 (5675)	405.4 (709)	320.8 (561)	299	0	100	27.2 ± 2.6
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MS 213	729721/9675308	15	48.35 (2913)	630.4 (819)	538.8 (700)	446	0	92	27.6 ± 3.0
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MS 390	725302/9693138	18	37.49 (3441)	318.1 (1054)	217.3 (720)	232	0	98	28.5 ± 3.0
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MS 281	720997/9667853	20	36.24 (3441)	548.6 (1339)	366.3 (1339)	404	0	95	28.4 ± 2.6
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MS 355	735230/9697240	1	39.58 (4173)	423.7 (114)	297.3 (80)	301	0	100	29.5 ± 4.6
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Appendix A (continued)

REGION, Formation, Sample	Grid reference	No. of grains analysed	$\rho_d \times 10^4 \text{cm}^{-2}$	$\rho_s \times 10^4 \text{cm}^{-2}$	$\rho_l \times 10^4 \text{cm}^{-2}$	U (ppm)	Var. (%)	$P(\chi^2)$ (%)	Age $\pm 2\sigma$ (Ma)	Depositional age if different from bulk age
QUINGEO AREA										
Chinchín Fm.										
MS 459	9680826/739467	7	40.92 (2792)	238.1 (400)	117.9 (198)	112	0	84	42.8 $\pm$ 3.8	
Quingeo Fm.										
MS 300	736413/9682577	11	37.94 (3441)	519.7 (893)	303.2 (521)	320	0	95	53.5 $\pm$ 10.0	34.9 $\pm$ 4.0
MS 412	728729/9664234	24	38.74 (3441)	542.7 (1719)	331.2 (1049)	333	0	89	35.9 $\pm$ 2.8	
MS 483	696981/9693135	19	41.03 (2792)	180.6 (1187)	103.4 (680)	98	0	98	37.1 $\pm$ 3.8	
MS 291	733101/9675508	16	35.55 (3441)	550.1 (1768)	271.3 (872)	298	0	99	37.4 $\pm$ 3.4	
MS 427	729933/9669540	20	38.92 (3441)	502.4 (1541)	264.4 (811)	272	0	100	38.7 $\pm$ 3.6	
MS 409	733464/9675553	13	37.26 (3441)	430.8 (838)	212.3 (413)	222	0	90	39.6 $\pm$ 5.0	
MS 410	732524/9671430	18	37.15 (3441)	491.5 (2615)	234.6 (1248)	246	6	21	161.1 $\pm$ 49.0	39.8 $\pm$ 2.8
MS 408	731963/9675027	20	37.83 (3441)	409.7 (1301)	203.8 (647)	216	2	53	39.8 $\pm$ 4.0	
MS 305	736675/9680785	20	36.58 (3441)	607.3 (1326)	275.7 (602)	302	0	95	42.2 $\pm$ 3.8	
GIRÓN AREA										
Santa Isabel Fm.										
DH 328	698813/9656909	12	30.60 (2541)	36.77 (94)	71.20 (182)	93.1	10	42	8.0 $\pm$ 2.2	
DH 333	699931/9642136	19	36.92 (2541)	428.1 (1295)	446.2 (1350)	484	17	1	17.6 $\pm$ 2.0	15.9 $\pm$ 1.6
Girón Fm.										
DH 330	702040/9644479	16	34.39 (2541)	107.7 (439)	184.3 (751)	214	0	74	10.1 $\pm$ 1.2	
DH 325	703293/9647227	17	41.98 (2541)	338.6 (742)	302.6 (663)	288	36	0	23.1 $\pm$ 5.0	10.3 $\pm$ 4.0
MS 240	703859/9647217	14	50.11 (6408)	562.9 (515)	659.1 (603)	513	6	37	21.6 $\pm$ 2.4	all detrital
Turi Fm.										
UH 142	683800/9633200	19	36.41 (2745)	31.51 (195)	79.34 (491)	87	0	90	7.4 $\pm$ 1.2	
DH 318	709399/9655972	6	35.15 (2745)	109.4 (227)	261.1 (542)	297	0	32	7.6 $\pm$ 1.2	
DH 334	703779/9649895	20	40.08 (2541)	69.61 (441)	156.1 (989)	156		90	9.0 $\pm$ 1.0	
MS 241	707940/9654581	4	54.53 (6408)	135.3 (130)	355.9 (342)	261	0	62	10.5 $\pm$ 2.2	
NABÓN AREA										
Saraguro Fm.										
MS 34	09820/25430	20	15.43 (1360)	295 (1322)	300 (1347)	175	37	1	26.7 $\pm$ 4.0	19.0 $\pm$ 3.5
DH 31	12250/29500	6	15.65 (1360)	295 (534)	275 (499)	207	8	26	28.2 $\pm$ 4.1	26.4 $\pm$ 4.5
DH 68	13480/31070	12	15.89 (1360)	331 (1054)	386 (1231)			41	22.9 $\pm$ 2.6	
MS 67	10450/27840	8	15.33 (1366)	293 (529)	337 (607)			9	22.5 $\pm$ 3.2	20.4 $\pm$ 3.1
MS 24	10365/24540	11	15.06 (1360)	174 (986)	201(1139)			11	22.0 $\pm$ 2.6	18.5 $\pm$ 6.0

Nabón Group											
WS 68	17650/31725	10	15.22 (1360)	685 (201)	223(653)	198	0	70	7.9 ± 1.4		
DH 92	16250/30880	12	14.83 (1090)	717 (261)	212(771)	152	7	39	8.5 ± 1.4		
DH 94	16220/30170	11	15.04 (1090)	534 (357)	144(963)			< 2	8.2 ± 1.3	7.5 ± 1.2	
WS 107	16900/34775	11	15.34 (1090)	609 (337)	176 (974)			58	9.0 ± 1.4		
DH 98	16300/28850	13	15.14 (1090)	124 (589)	319(1512)			< 2	11.0 ± 2.0	8.3 ± 1.1	
MS 105	11110/27870	9	13.93 (1090)	156 (522)	343(1148)			8	10.7 ± 1.4	8.9 ± 1.3	
Tarqui Fm. (Tambo Viejo Mb.)											
MS 100	13550/28200	10	15.44 (1090)	789.0 (283)	325.0 (1164)	223	0	73	6.3 ± 1.0		
SANTA ISABEL AREA											
Saraguro Fm.											
MS 237	681315/9630791	18	45.86 (5787)	618.8 (2086)	697.7 (2352)	609	8	13	20.3 ± 1.6	19.1 ± 1.4	
CH 61	688400/9629400	3	40.20 (2745)	241.9 (93)	236.7 (91)	236	0	78	21.1 ± 6.2		
DH 464	671534/9632381	7	51.56 (2745)	575.2 (398)	721.2 (499)	560	0	81	21.1 ± 3.0		
DH 486	697035/9625187	28	49.03 (2745)	349.5 (1228)	354.9 (1247)	282	16	3	25.2 ± 2.8	23.0 ± 2.2	
DH 487	697311/9624303	20	45.88 (2745)	378.9 (1554)	381.3 (1564)	332	0	88	23.4 ± 2.0		
CH 66	680200/9630300	15	49.96 (3020)	474.4 (972)	452.4 (927)	362	0	92	26.4 ± 2.6		
Santa Isabel Fm.											
MS 239	683296/9633982	24	48.64 (6408)	142.6 (912)	189.8 (1214)	152	1	59	18.4 ± 1.6		
CH 62	688300/9629400	16	55.73 (3020)	276.4 (780)	419.6 (1184)	301	6	58	18.4 ± 2.0		
DH 483	689714/9631727	8	43.98 (2745)	527.6 (576)	560.6 (612)	510	14	7	21.1 ± 3.4	18.4 ± 2.8	
DH 339	698428/9640804	15	39.14 (2541)	377.9 (600)	396.8 (630)	406	0	92	18.8 ± 2.2		
Burrohuaycu Fm.											
UH 61	687200/9634700	17	39.56 (2745)	55.85 (292)	108.5 (567)	110	0	100	10.5 ± 1.6		
UH 141	685100/9635600	17	37.04 (2745)	70.44 (352)	124.5 (622)	134	0	99	10.8 ± 1.4		
DH 480	684834/9635468	20	46.51 (2745)	55.57 (370)	121.5 (809)	105	0	93	10.9 ± 1.4		
DH 398	687750/9630765	19	37.55 (2541)	86.94 (513)	142.2 (839)	152	8	53	11.6 ± 1.4		
UH 82	685100/9635600	15	53.63 (3020)	54.40 (188)	126.2 (436)	94	0	97	11.7 ± 2.0		
CH 68	680300/963000	15	51.53 (3020)	66.11 (231)	138.5 (484)	108	2	82	12.4 ± 2.0		
CH 63	688100/962900	8	50.75 (3020)	62.10 (132)	126.1 (268)	99	0	58	12.6 ± 2.8		
CH 118	684300/9626900	11	38.93 (2745)	133.5 (354)	209.6 (556)	215	0	100	12.8 ± 1.8		
DH 337	681205/9630156	9	50.30 (2745)	62.87 (203)	114.9 (371)	91	0	92	14.2 ± 2.6		
MS 238	683296/9633982	19	52.81 (6408)	206.4 (1087)	374.3 (1971)	284	8	13	14.7 ± 1.2		
DH 335	685570/9636150	14	35.34 (2541)	153.0 (327)	173.1 (370)	191	1	73	15.8 ± 2.4	all detrital	
DH 415	681406/9628402	10	32.49 (2541)	195.7 (474)	170.5 (413)	205	0	85	18.8 ± 2.6	all detrital	

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Appendix A (continued)

REGION, Formation, Sample	Grid reference	No. of grains analysed	$\rho_d \times 10^4 \text{cm}^{-2}$	$\rho_s \times 10^4 \text{cm}^{-2}$	$\rho_l \times 10^4 \text{cm}^{-2}$	U (ppm)	Var. (%)	$P(\chi^2)$ (%)	Age $\pm 2\sigma$ (Ma)	Depositional age if different from bulk age
Uchucay Fm.										
MS 236	682810/9629759	28	46.07 (5787)	190.4 (1866)	221.6 (2172)	192	47	0	18.9 $\pm$ 1.7	9.4 $\pm$ 1.6
LOJA AREA										
Loma Blanca Fm.										
MS 225	699139/9546868	8	42.41 (3172)	518.3 (522)	302.8 (305)	286	17	6.7	36.2 $\pm$ 6.8	
San Cayetano Fm. (Lower Sandstone Mb.)										
DH 246	701249/9559170	23	46.45 (5787)	226.1 (962)	285.2 (1639)	323	0	88	13.8 $\pm$ 1.2	
San Cayetano Fm. (Siltstone Mb.)										
MS 226	699052/9547885	13	38.29 (3172)	130.5 (321)	253.2 (623)	265	0	86	10.0 $\pm$ 1.4	
DH 375	698987/9563773	20	44.02 (3020)	321.0 (1240)	635.2 (2454)	575	7	12	11.4 $\pm$ 1.0	10.6 $\pm$ 0.8
DH 247	699147/9548622	12	37.26 (3172)	98.82 (321)	173.6 (564)	186	0	91	10.7 $\pm$ 1.6	
La Banda Fm.										
DH 223	696922/9561374	16	40.35 (3172)	242.7 (737)	446.2 (1355)	442	9	22	11.1 $\pm$ 1.2	
Salapa Fm.										
DH 225	693545/9569306	23	46.36 (5787)	431.0 (1327)	431.6 (1329)	363	105	0	16.4 $\pm$ 7.4	2.3 $\pm$ 0.8
MALACATOS–VILCABAMBA AREA										
Tres Lagunas Metagranite										
MS 254	692092/9538900	26	51.33 (5668)	665.3 (2606)	278.5 (1091)	217	7	35	61.6 $\pm$ 5.0	
Loma Blanca Fm.										
DH 344	685992/9538147	12	45.77 (2541)	237.0 (431)	205.7 (374)	175	0	97	26.6 $\pm$ 4.0	
DH 314	685900/9538200	13	52.80 (2745)	876.0 (1219)	720.1 (1002)	532	3	58	33.0 $\pm$ 3.2	
DH 233	687184/9537098	13	45.52 (5787)	111.1 (854)	763.5 (587)	671	0	83	33.4 $\pm$ 3.6	
MS 247	700498/9519437	24	45.73 (5787)	555.8 (1410)	244.4 (620)	208	249	0	50.5 $\pm$ 6.4	40.6 $\pm$ 5.4
Solanda Mb.										
DH 343	686261/9537034	20	47.04 (2541)	294.5 (841)	191.9 (548)	163	7	74	36.5 $\pm$ 4.4	
Purunuma Quartzporphyry Mb.										
DH 235	683017/9535111	17	48.84 (5668)	460.8 (1107)	374.6 (900)	307	0	64	30.3 $\pm$ 2.4	
Quinara Fm.										
DH 291	697477/9526770	8	50.83 (2541)	85.42 (110)	154.5 (199)	119	0	98	14.2 $\pm$ 3.4	
MS 230	694649/9523782	25	53.79 (6408)	130.6 (637)	237.6 (1159)	172	3	54	14.9 $\pm$ 1.6	
MS 228	694649/9523782	17	49.13 (6408)	156.6 (669)	255.9 (1093)	203	7	71	15.1 $\pm$ 1.6	
San José Fm.										
MS 250	696247/9534744	20	42.74 (5668)	184.9 (425)	272.3 (626)	249	15	17	14.6 $\pm$ 2.2	13.1 $\pm$ 1.8
DH 232	688664/9536727	16	48.08 (5668)	194.5 (924)	351.7 (1671)	285	5	43	13.5 $\pm$ 1.2	
Santo Domingo Fm.										
MS 255	695648/9536185	20	46.17 (5668)	110.8 (666)	214.4 (1288)	181	4	46	12.0 $\pm$ 1.2	
DH 317	696345/9532381	7	42.61 (2541)	240.0 (369)	420.1 (646)	394	0	93	12.3 $\pm$ 1.6	

MS 245	689883/9541970	12	47.13 (5668)	320.7 (620)	551.3 (1066)	468	61	0	14.8 ± 5.4	12.4 ± 1.0
MS 252	694924/9536436	25	54.04 (6408)	256.3 (1121)	495.6 (2168)	358	18	0	14.0 ± 1.6	13.2 ± 1.4
DH 228	689836/9536663	24	43.88 (5668)	149.4 (626)	227.9 (955)	202	10	31	14.5 ± 1.6	14.1 ± 1.6
DH 229	689804/9536110	10	50.37 (5668)	204.4 (389)	356.3 (678)	283	0	63	14.6 ± 1.8	
Cerro Mandango Fm.										
DH 294	697791/9525385	20	48.30 (2541)	700.8 (586)	221.5 (1852)	179	0	66	7.7 ± 0.8	
DH 293	697791/9525385	15	49.57 (2541)	827.5 (369)	259.7 (1158)	210	0	91	8.0 ± 1.0	
DH 292	697250/9526717	20	41.05 (3020)	120.6 (909)	212.1 (1598)	207	12	13	11.8 ± 1.2	10.0 ± 1.0
DH 300	695180/9524530	7	44.51 (2541)	978.6 (120)	209.6 (257)	188	0	99	10.5 ± 2.4	
MS 253	693453/9537784	19	41.97 (5668)	220.0 (553)	369.9 (930)	344	36	0	12.6 ± 2.6	11.0 ± 1.6
DH 373	690708/9540396	20	43.88 (2541)	107.3 (564)	212.0 (1115)	193	0	99.0	11.2 ± 1.2	
MS 248	695240/9531885	19	35.71 (3172)	449.1 (183)	721.5 (294)	81	0	76.8	11.2 ± 2.2	
MS 227	697217/9527770	21	39.30 (5668)	728.8 (411)	117.6 (663)	117	16	29.3	12.0 ± 1.8	11.4 ± 1.6
CATAMAYO–GONZANAMÁ AREA										
Rodanejo Pluton										
DH 450	672930/9550569	16	48.40 (2745)	360.5 (661)	232.9 (427)	193	12	22	38.7 ± 5.6	
El Tingo Pluton										
DH 451	687422/9558650	10	40.83 (2745)	272.0 (551)	269.5 (546)	264	0	94	21.2 ± 2.6	
Loma Blanca Fm.										
MS 259	678920/9562364	12	383.9 (5675)	417.9 (604)	320.3 (463)	325	0	98	25.2 ± 3.2	
DH 391	680590/9544440	20	43.15 (3020)	293.2 (1465)	218.1 (1090)	202	9	20	29.0 ± 2.8	
Gonzanamá Fm.										
DH 443	675103/9540573	10	42.09 (2745)	175.3 (190)	232.5 (252)	215	19	15	16.4 ± 3.8	14.0 ± 3.0
DH 394	674378/9539250	20	42.10 (3020)	766.9 (523)	113.2 (772)	108	1	80	14.4 ± 1.8	
DH 439	674378/9539250	17	43.35 (2745)	155.4 (506)	221.1 (720)	204	1	78	15.7 ± 2.0	
PLAYAS AREA										
Sacapalca Fm.										
DH 385	646051/9555116	25	48.39 (3020)	643.3 (2903)	233.1 (1052)	193	6	40	66.9 ± 5.8	
Loma Blanca Fm.										
DH 387	646676/9558934	20	46.82 (3020)	454.6 (1295)	344.8 (982)	295	0	88	31.1 ± 2.8	
DH 388	647582/9559391	20	45.77 (3020)	619.1 (2380)	337.9 (1299)	295	4	58	42.2 ± 3.4	

For details of methods see Hungerbühler et al. (1995), Steinmann (1997), Hungerbühler (1997) and Steinmann et al. (1999).

Grid references are in Universal Transverse Mercator (UTM) form, except for the Nabón Basin where the co-ordinates are taken from local topographic maps (Instituto Geográfico Militar, 1970, 1971).

ρ_d , ρ_s and ρ_i represent track densities in the dosimeter, and the densities of spontaneous and induced tracks in the target mineral, respectively. Numbers in parantheses represent tracks counted. $P(\chi^2)$ is the probability of obtaining χ^2 values for ν degrees of freedom where ν = number of crystals – 1.

Ages are calculated using the zeta approach (Hurford and Green, 1983). Zircon-SRM 216 zeta for Nabón samples was 338 ± 5 , for the Cuenca and Quingeo Basins zeta for zircon-CN1 was 105 ± 2 , and for the remaining zircons the CN1 was 103 ± 5 . All ages are central ages (Galbraith and Laslett, 1993); errors are calculated according to Green (1981) and are expressed at the 2σ level. Where several age components could be resolved, the youngest population is taken to represent the time of sedimentation (see method in Steinmann et al., 1999). These ages are listed in the last column.

In addition to the Cordillera Real, the newly rising Cordillera Occidental in the west and the Huancabamba Andes (related to the activity of the Jubones clockwise rotational fault) in the south provided clastic material to the Intermontane basins (Fig. 18C). However, coeval displacement of the Costa and Cordillera Occidental block along the Calacali–Palatanga Fault zone resulted in strong subsidence in the Jambelí–Tumbez pull-apart basin.

The period of basin inversion at around 10–9 Ma is well constrained by facies development and the ages of unconformities defined by ZFT analyses. Further evidence is provided by modelling of apatite fission-track data from the thick Cuenca Pacific Coastal sequence, which suggests that exhumation has occurred since ≈ 9.5 Ma until present (Steinmann, 1997; Steinmann et al., 1999). Coeval rapid exhumation was occurring in the northern Cordillera Real (Spikings et al., 2000, 2001), indicating that compression also involved the Andean chain of northern Ecuador. Refined plate reconstructions (Spikings et al., 2001) suggest that the geologic history of the southern Ecuadorian forearc and arc was driven by the collision of the Carnegie Ridge with the Ecuadorian margin some time between ≈ 15 and 9 Ma. The tectonic and sedimentary development in the forearc and arc, however, points more strongly to a collision at ≈ 15 Ma.

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Dominik Hungerbühler graduated in geology at the University of Zürich in 1993. His research in southern Ecuador earned a PhD in geology in 1997 from the ETH Zürich. He has been working since 1997 with Shell International Exploration and Production and is currently on assignment in the Netherlands as a Production Geologist. His current work is focussed on the development of gas and oil fields in the Dutch North Sea sector. His research interests lie still

in the geodynamic evolution of the Andes, and the static and dynamic reservoir modelling.



Michael Steinmann graduated in 1993 at the University of Zurich. In 1997 he obtained a PhD from ETH Zürich in geology for his work in basin analysis and tectonic reconstructions in Ecuador. Then he worked as a mining consultant in Chile and Ecuador. He is currently Chief Geologist for Glencore Operations in Peru with additional duties in Turkey. His main research interests are the genesis and new exploration methods of base metal ore deposits especially along convergent plate margins.



Wilfried Winkler has held since 1988 a position of research associate and senior lecturer for sedimentary petrology, basin analysis and sedimentology at the ETH Zürich. He graduated in 1977 at University of Fribourg (Switzerland) where he subsequently obtained a PhD in geology in 1981. From 1981–1988, he was a postdoctoral fellow and lecturer at both Fribourg and Basel Universities. His main research interests are the bearings of tectonics and climate on basin sedimentation and provenance studies in different plate tectonics settings. Whilst his earlier research was focussed on Alpine, Carpathian and Pyrenean turbidite (flysch) and mélange formations, since 1991 he co-leads several Master's, PhD and Postdoctoral projects in the Andes of Ecuador and Peru.



Diane Seward is a senior scientist at the ETH Zürich (Switzerland). She obtained degrees from UC Wales, Aberystwyth (BSc Hons), McMaster University, Canada (MSc) and a PhD at University of Victoria, Wellington, New Zealand. Later she held a postdoctoral fellowship in Wellington, and was a visiting scientist at the Max Planck Institut für Kernphysik, Heidelberg, Germany. She then worked for the DSIR, New Zealand. Since 1990 she has been at the ETH Zürich. Her interests lie in the use of fission track analysis and other low temperature techniques to answer the many and varied questions related to different aspects of geology.



Arturo Egüez Delgado has been since 1982 professor for geology at the Escuela Politécnica Nacional (EPN) in Quito-Ecuador. In 1986 he obtained a PhD from Paris IV University (France) for his work on the structures and metallogeny of the Western Cordillera of Ecuador. He is a consulting geologist and co-author of several geological, tectono-metallogenic and seismotectonic national maps of Ecuador. His current subjects of research are Andean regional geology, tectonics and metallogeny.



Dawn Peterson is a curatorial assistant and research associate affiliated with the California Academy of Sciences in San Francisco, California. Due to a medical condition, she is self-educated through practical application in the field of ostracodology. She is currently engaged in research with the Panama Paleontology Project, researchers and graduates of the Swiss Federal Institute of Technology (ETH-Zürich), and the National Natuurhistorische Museum, Leiden, the Netherlands,

studying ostracode faunas of mid-Miocene southern Ecuador and Pebasian Peru. Additionally she is assembling collections of Recent and paleontological eastern Pacific ostracode faunas for the California Academy.



Urs Helg graduated in geology in 1997 at the ETH Zürich. Currently, he is at the Geological Institute of Neuchâtel University (Switzerland) finishing his PhD thesis on the structural geology and tectonics of the Hercynian orogeny in the Anti-Atlas of southern Morocco. His fields of interest are inversion tectonics, basin inversion and low grade metamorphism.



Cliff Hammer obtained in 1998 a MSc degree in geology from ETH in Zürich. After a 9-month travelling experience and insight to the human resources sector of a telecommunication enterprise, he obtained in 2000 a MSc degree in hydrogeology from the Centre de Hydrogéologie (Chyn) at the University of Neuchâtel, Switzerland. He is currently working as a consulting hydrogeologist in the area of environmental engineering, dealing mainly with remediation of contaminated sites and groundwater risk assessment.